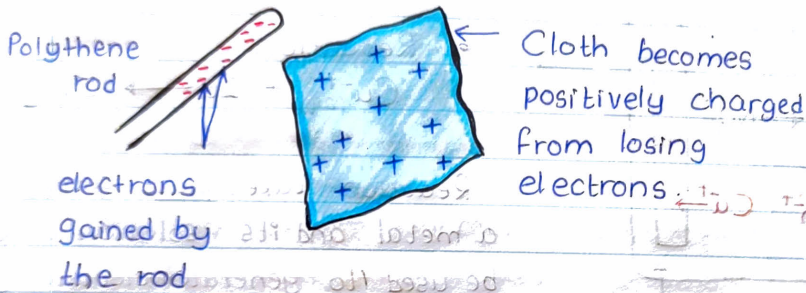


THE ELECTRICAL NATURE OF MATTER

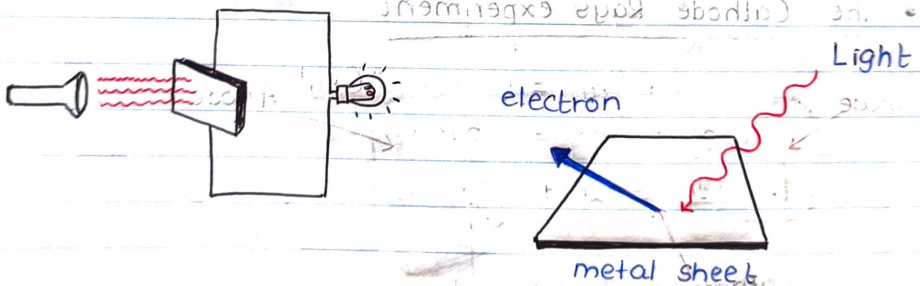
PHYSICAL EVIDENCES

- ① The production of static electricity upon the friction of two matters.



- ② Photo-Electric Effect

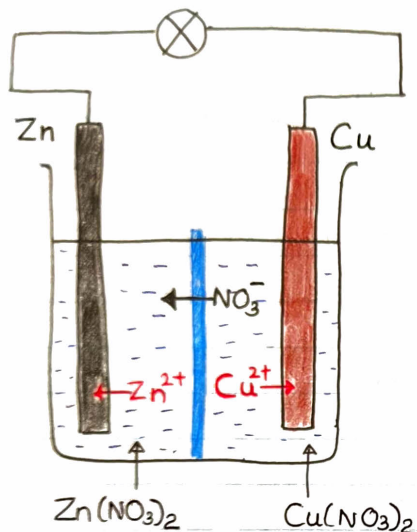
When light of short wave-length is allowed to shine the surface of some metals, which produce electric electricity.



- ③ When a gas (Hg vapour) is subjected on higher potential under low pressure (0.01 mm Hg) the wall of the glass tube fluoresce with green colour.

CHEMICAL EVIDENCE

* Galvani Cell



• Oxidation



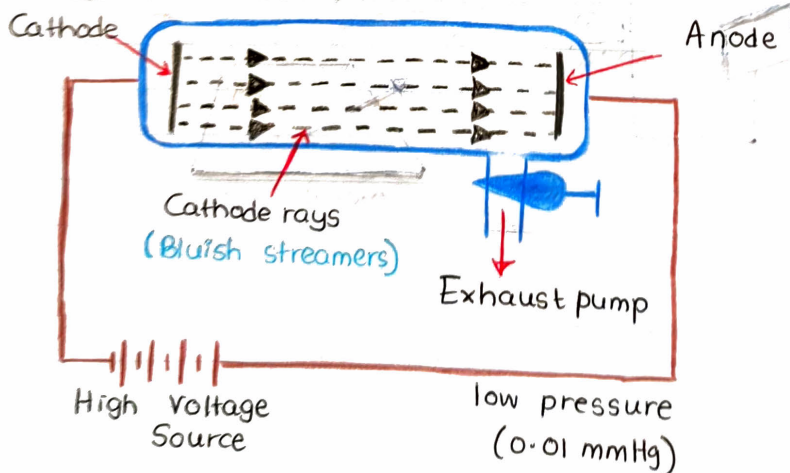
• Reduction



Reaction takes place between a metal and its solution could be used to generate electricity.

THE EXISTENCE OF ELECTRON

• The Cathode Rays experiment

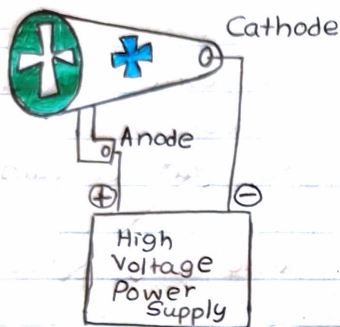


* Observed by → William Crooke

* Named as Cathode rays by → Goldstein

• Properties of Cathode Rays

- ① Travel in straight lines, cast shadows of objects placed in their path.

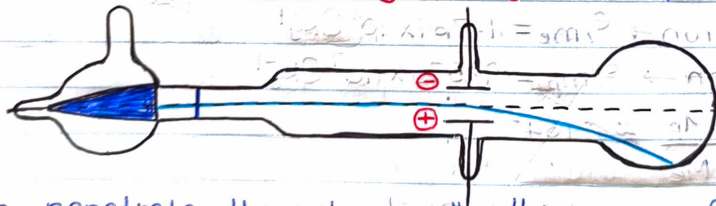


Electron

- A universal particle
- e/m ratio was discovered by J.J. Thompson ($1.759 \times 10^{11} \text{ C/kg}$)
- Charge of the electron ($1.602 \times 10^{-19} \text{ C}$) was accurately determined by Robert Millikan $\rightarrow$ Oil drop experiment

- ② Exert mechanical pressure.
(When rays are incident on one of the vane surfaces of a rotating wheel, the wheel is set in motion. They possess momentum)

- ③ Deflected by magnetic or electric field.
(as they are negatively charged $(-)$ particles)



- ④ Can penetrate through small thickness of matter.

Ratio of e/m of Cathode rays particles doesn't depend on,

- ① Nature of gas
- ② Applied pressure
- ③ Applied potential
- ④ Nature of the Cathode and Anode material.

POSITIVE RAYS

- Properties of positive rays
 - ① Move in straight line.
 - ② Have a momentum hence particle nature
 - ③ Deflected by an electric field opposite to that of Cathode rays, hence positively charged.
 - ④ e/m ratio value depends on the gas contains in the discharge tube.
 - ⑤ The highest e/m ratio of $9.575 \times 10^7 \text{ Ckg}^{-1}$ gives for H_2 in the tube.
 - ⑥ Mass $\approx$ Mass of H atom
 - ⑦ For other gases, the mass of positive particle is whole number multiples of the mass of H^+ particle.
 - ⑧ This particle is also common to matter and named as proton.

A contrast between electron and proton

- ① For electron $\rightarrow e/m_e = 1.769 \times 10^{11} \text{ Ckg}^{-1}$
For proton $\rightarrow e/m_p = 9.575 \times 10^7 \text{ Ckg}^{-1}$
- ② $\frac{M_p}{M_e} = 1847$

③ $M_p = 1847 \times 9.107 \times 10^{-31} \text{ Ckg}^{-1}$
 $\therefore M_p = 1.673 \times 10^{-27} \text{ kg}$

- ⊗ The mechanism of the formation of positive particles in the discharge/cathode ray tube.

In the discharge tube a small number of residual gas molecules are contained between the cathode and the anode. The electrons that are emitted from the cathode collide with the residual gas molecules. These collisions are not an electron out of each gaseous atom generating positively charged ion which are accelerated towards the cathode and they pass through the holes in the cathode producing a beam of positively charged particles of the particles of the the back of the cathode. As the positively charged particles pass through the holes or the canals in the cathode they are called Canal rays.

DISCOVERY OF NEUTRON

- The mass of H atom = sum of the mass of p and e^-
For higher mass number elements this does not hold fit

Mass of the particular atom $> n_1 m_p + n_2 m_e$

Therefore, the extra mass is described to be presence of another uncharged particle.

- James Chadwick proved the presence of the particle and named as Neutron.
- In the nucleus there can't exist only the protons as like charges get repelled. Neutrons bind mutually repelling protons together. In the nucleus there should be some sort of attractive force which is strong enough to overcome the strong repulsive Coulombic force acting between protons.

attractive forces act only at very short range (not more than 10^{-15} m)

EXPERIMENTS FOR ATOMIC STRUCTURE

ALPHA-PARTICLE SCATTERING EXPERIMENT

- Carried out by → Geiger and Marsden
- Final conclusions were made by → Ernest Rutherford

Observations	Conclusions
① Most of the alpha particles passed straight through the foil.	Most of the space of the atom occupied by the electron is empty space.
② Some of the alpha particles are deflected through large angles.	They must have approached some positively charged region with a heavy mass.
③ A very few alpha particles experienced deflection backwards.	Volume occupied by the nucleus is very small. Almost the complete mass of the atom must be within the nucleus.

- Shortcomings of Rutherford's Model

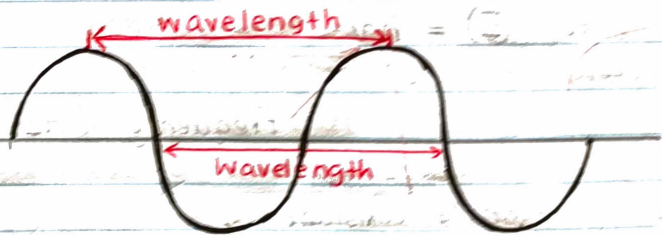
⊛ According to electromagnetic theory, an accelerated electrical charge should lose its energy in the form of e-m-r. Due to the fact, the electron should emit radiation and lose energy as electromagnetic radiation and spiral down to the nucleus. But the atoms under normal circumstances do not radiate energy. Rutherford gave an concept of unstable atom. ⊛ And also, he didn't say anything about the arrangement of electrons in the atom.

ATOMIC SPECTRA OF HYDROGEN - BOHR MODEL OF THE ATOM

The low pressure Hydrogen gas shows a characteristic spectra upon an electric discharge and found a series of lines called emission spectrum of Hydrogen.

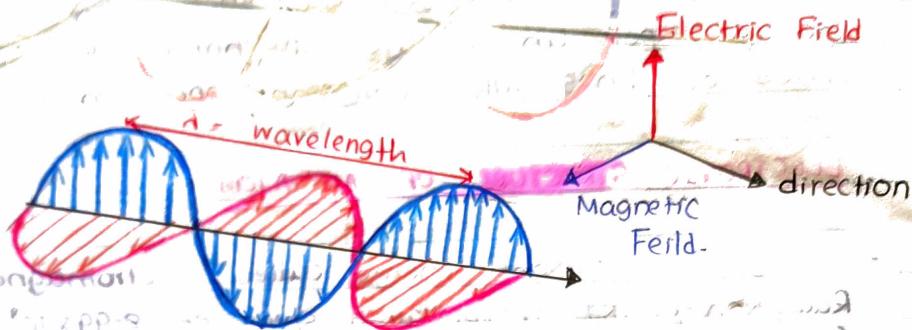
ELECTRONIC STRUCTURE OF AN ATOM

- All types of radiant energy called Electromagnetic waves Radiation (EMR) travel at a speed of $2.99 \times 10^8 \text{ m/s}$ ($\approx 3 \times 10^8 \text{ m/s}$)
- James Maxwell showed that all known properties of light could be accounted for by means of an equation based on the hypothesis that light is an electromagnetic wave.



- * wavelength $\rightarrow$ The distance between successive peaks (or troughs)
- * Frequency $\rightarrow$ The number of complete wavelengths that pass a given point in one second

- * EMR consist of electric and magnetic fields that oscillate in direction perpendicular to each other.



- * Einsten explained that the light was not to be regarded as a continuous electromagnetic field, but was a made up of discrete packets of energy known as photons.

Each photon has an energy (E)

very tiny packet
of energy or
quanta

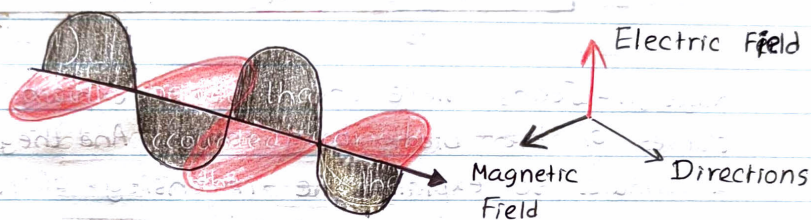
$$(E) = h\nu$$

Frequency (Hz or s^{-1})

Plank's constant
 $6.6 \times 10^{-34} \text{ J s}^{-1}$

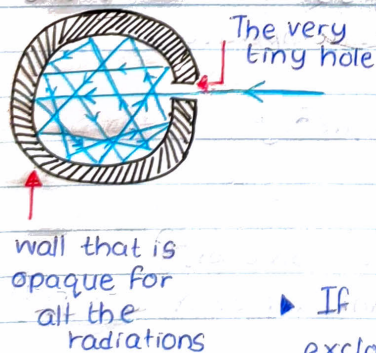
BASIC CONCEPTS

- * Anything that has a mass and occupy (the space) a volume is known as a matter.
- * All matters above absolute zero (according to the Kelvin scale) emits or absorb electromagnetic radiations. (Electromagnetic radi waves are waves that have both an electric component and a magnetic component)



- * A matter that can absorb all the radiations with all the frequencies and or emit the radiations with all the frequencies is known as a black body. (idealized case)

Approximate example → Sun

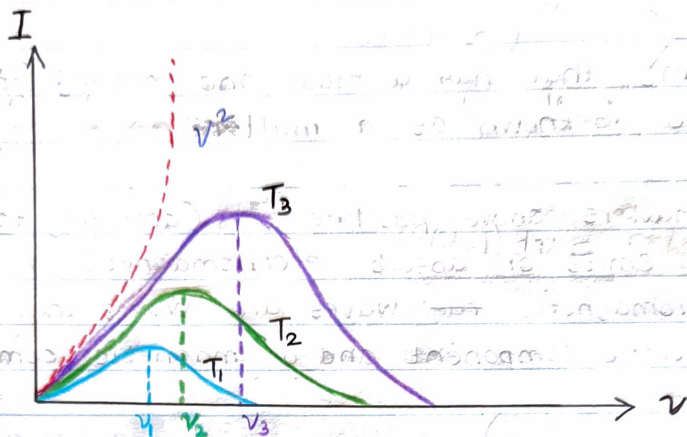


► Any radiation that enters the blackbody through the tiny hole can undergo a repeated reflection (i.e. all the energy of the radiation is absorbed to the walls of the black body).

► If the radiations ^{are} exceeded in the enclosure it starts to emit radiations.

- Its emission has a characteristic frequency distribution. This depends on the temperature. (only)

* Radiation intensity against wave frequency.



Rayleigh - Jeans were the first to explain experimental curves of the black body range. And they developed a formula to explain the intensity of radiation as a function of frequency.

According to the relative law they found

$$\rho(\nu, T) d\nu = \frac{8\pi k_b T}{C^3} \nu^2 d\nu$$

density of radiation energy between ν and $(d\nu + \nu)$ at T (a particular temperature)

small changing of the frequency

ν - frequency

C - velocity of light ($3 \times 10^8 \text{ ms}^{-1}$)

k_b - Boltzman Constant ($1.38 \times 10^{-23} \text{ JK}^{-1}$)

- * According to their law they found that the curve of Intensity vs. Frequency rises to the infinite
- * For the higher frequencies the ^{divergence} deviation of the intensity vs. frequency curve is over the experimental curve. This divergence of at higher frequencies are called ultra violet catastrophe.

* They assumed that

In the derivation of Rayleigh-Jeans law, they assumed that, radiation is due to the vibrations of electronic vibration oscillation in the material and continuous.

* 1900 a scientist called Max Planck made a revolutionary assumption in the derivation of a formula which could not account for the experimental observations.

* The hypothesis that Max Planck used to explain blackbody mechanism. It is known as Planck's Quantum Hypothesis.

* Max Planck proposed that the radiation absorbed or emitted in certain discrete amounts ~~cat~~ is known as a quantum.

* Energy of the electronic oscillator is an integral multiple of a minimum energy

$$E = n E_{\text{minimum}}$$

$$E \propto \nu$$

$$E = h\nu$$

$$E = h\nu$$

h - Planck constant

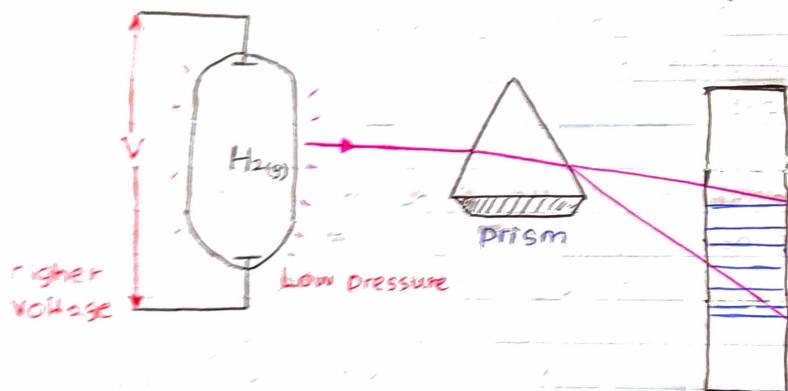
$$\rho(\nu, T) d\nu = \frac{8\pi h \nu^3}{c^3} \frac{1}{\exp(h\nu/k_B T) - 1}$$

Planck's distribution Law

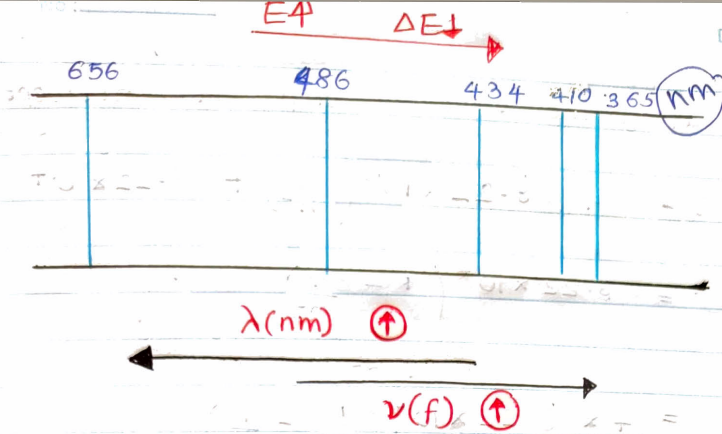
$$h = 6.63 \times 10^{-31} \text{ Js} \quad (\text{in } 1900)$$

- Five years later Albert Einstein came up with the idea of photo electric effect.

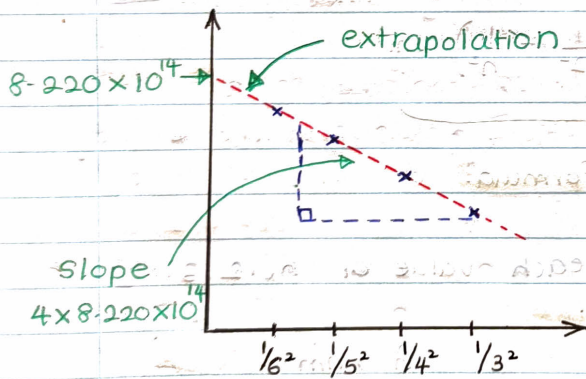
Day 03 / 22-01-2022



- When a high voltage is applied for Hydrogen gas placed in a tube at a reduced pressure, it emits lights.
- Different gases emit different colours of light.
- When the light coming from such tube is dispersed by a prism, only certain wavelengths are found to be present. (And also this is discontinuous spectrum)
- A spectrum containing radiation only specific wavelengths is called a line spectrum.
- Also known as the emission spectrum of the hydrogen gas.



- The distance between any two consecutive lines increases with the wave length.
- The distance between any two consecutive lines decreases with the frequency.
- ^{Schumann} Balmer was the one who tried to build up a relationship between line frequencies. He plotted frequency of each line against a certain ($1/n^2$)



- Balmer found a straight line relationship between ν vs. $1/n^2$.

$$y = mx + c$$

ν $\rightarrow$ y $\quad$ $1/n^2$ $\rightarrow$ x

- By substituting the value of slope and intercept

$$\nu = -4 \times 8.22 \times 10^{14} \left(\frac{1}{n^2} \right) + 8.22 \times 10^{14}$$

$$= -8.22 \times 10^{14} \left(1 - \frac{4}{n^2} \right)$$

$$= 4 \times 8.22 \times 10^{14} \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

$$c = \lambda \nu$$

$$\frac{1}{\lambda} = \frac{\nu}{c}$$

$$\frac{1}{\lambda} = \bar{\nu} = \frac{\nu}{c} = \frac{4 \times 8.22 \times 10^{14}}{3 \times 10^8} \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

$$2.994 \times 10^6 \text{ m}^{-1}$$

$$\frac{1}{\lambda} = \bar{\nu} = 1.0968 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \text{ m}^{-1}$$

wave number $\rightarrow \bar{\nu} = 109680 \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \text{ cm}^{-1}$

Balmer Formula

He calculated $\bar{\nu}$ for each value of $n(2, 3, \dots)$

n	$\bar{\nu} / \text{cm}^{-1}$	λ / nm
3	15233	656.4
4	20565	486.3
5	23032	434.2
6	24373	410.2

These values were close to agreement with experimental value.

- for very large values of n , $\frac{1}{n^2} \rightarrow 0$
 $\therefore \bar{\nu} = 109680 \left(\frac{1}{2^2} - 0 \right)$
 $= 27420 \text{ cm}^{-1}$
 $\lambda = 364.7 \text{ nm}$

- Balmer equation given a relationship between spectral frequencies of Hydrogen emission spectrum in UV visible.
- But the emission spectrum consist of several lines in the different region of E.M.R. spectrum. These regions are named as follows.

Series	Region of E.M.R.	n_1	n_2
Lyman	UV	1	2, 3, 4, ...
Balmer	near UV-Vis	2	3, 4, 5, ...
Paschen	near IR	3	4, 5, 6, ...
Brackett	IR	4	5, 6, 7, ...
Fund	far IR	5	6, 7, 8, ...

- Johann Rydberg introduced an empirical formula for all spectral series.

$$\bar{\nu} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad n_2 > n_1$$

R_H is known as Rydberg constant = $2.18 \times 10^{-18} \text{ J}$
 or 109680 cm^{-1}

- Rydberg formula demonstrates that the existence of electron energy levels as $n = 1, 2, 3, \dots$ and that indicates some sort of quantization of energy levels. It was Neils Bohr picked up this idea of quantization to theoretically explain the Rydberg formula and thereby emission spectra of (H) atoms.

Bohr Model of atom

- Bohr used Planck's quantum theory to explain the line spectrum of Hydrogen.

Postulations of the atom

- Negatively charged electrons are revolving in circular orbits known as stationary energy levels. An electron in an allowed stationary energy state will not spiral into the nucleus.

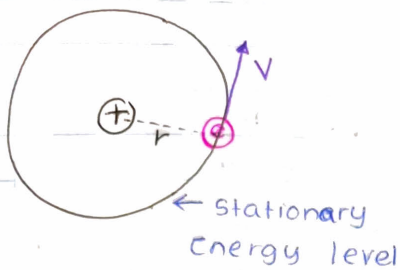
- Angular momentum of electron is quantized by $(\frac{h}{2\pi})$

$$mvr = n\left(\frac{h}{2\pi}\right) \quad \text{--- (1)} \quad n = 1, 2, 3, \dots \text{etc}$$

Atoms emit light when they are heated or excited at higher energy levels. When the electrons return to low energy levels, they release extra energy and that can be in the form of light.

causing the emission of light. On the other hand absorb light is light that isn't seen.

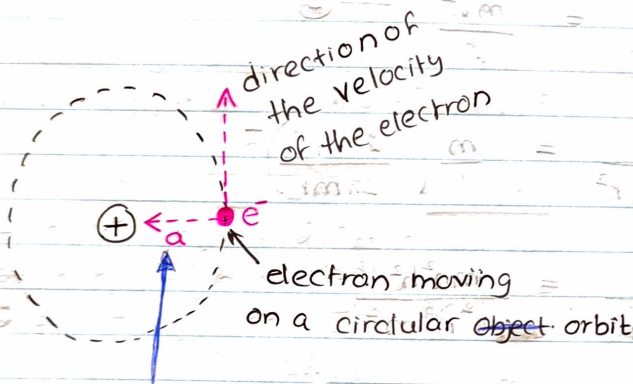
- $m \rightarrow$ mass of the electron.
- $v \rightarrow$ velocity of the electron.
- $r \rightarrow$ radius of the orbit.
- $h \rightarrow$ Planck's constant
- $n \rightarrow$ quantum numbers.



③ Electron doesn't emit radiation as long as it is in the a particular fixed orbit energy level

④ Electrons emit or absorb energy only when they transfer between two energy levels.

⑤ Consider an electron moving around a positively charged nucleus in an orbit of radius-



Centripetal force $\rightarrow \frac{mv^2}{r}$ this always directs towards the centre of the orbit.

* This force comes from the electrostatic interaction between the positive nucleus and the negative electrons.

* The basic force law of electrostatic is,

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$$

← charges
↑ dielectric permittivity of free space.

* electrostatic attraction

$$F = \frac{e^2}{4\pi\epsilon_0 r^2}$$

but

$$mvr = \frac{nh}{2\pi} \quad \text{--- (1)}$$

$$v = \frac{nh}{2\pi mr}$$

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r} \quad \text{--- (2)}$$

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{m}{r} \left(\frac{nh}{2\pi mr} \right)^2$$

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{mn^2h^2}{4\pi^2 r^3 m}$$

$$e^2 = \frac{\epsilon_0 n^2 h^2}{\pi m r^2}$$

radius of the Hydrogen atom $\rightarrow r =$

$$\frac{\epsilon_0 n^2 h^2}{\pi m e^2} \quad \text{constant}$$

∴ So, that can be written as,

$$r = n^2 \cdot \text{constant} \quad \text{--- (3)}$$

- ⑥ * Equation suggests that ⑦ can have only certain values ($r = 1^2 \cdot \text{constant}$, $r = 2^2 \cdot \text{constant}$, $r = 3^2 \cdot \text{constant} \dots$)
Then reveals an electron can move only in certain radius.

- * For its atom radius of the 1st orbit is,

$$r_1 = \frac{n^2 \epsilon_0 h^2}{\pi m e^2}$$

$$n=1 \therefore r = 5.29 \times 10^{-11} \text{ m}$$

- * This is ~~now~~ ~~Birr~~ known as the Bohr radius

- * Let's consider the energy of an electron in the orbit
The energy is partly potential and kinetics.

Total Energy

$$E_T = E_p + E_k$$

$$= \frac{-e^2}{4\pi\epsilon_0 r} + \frac{1}{2}mv^2 \quad \text{--- ④}$$

From equation ②

$$\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

$$mv^2 = \frac{e^2}{4\pi\epsilon_0 r}$$

$$E_T = \frac{1}{2} \frac{e^2}{4\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r}$$

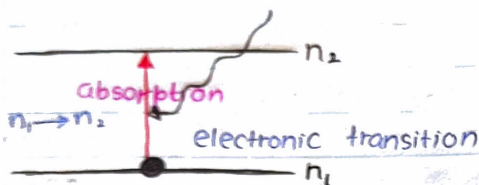
$$= \frac{-e^2}{4\pi\epsilon_0 r} \times \frac{\pi m e^2}{\epsilon_0 n^2 h^2} \cdot \frac{1}{r}$$

$$E_T = \frac{-m e^4}{8 \epsilon_0^2 n^2 h^2}$$

$$= \left(\frac{-1}{n^2} \right) \times \text{constant}$$

Since $n = 1, 2, 3 \dots$ etc.
the total energy is quantized.

⑦ Negative sign of energy is in Bohr orbit indicates less energy



$$E_2 = \frac{-me^4}{8\epsilon_0^2 n_2^2 h^2} \quad E_1 = \frac{-me^4}{8\epsilon_0 n_1^2 h^2}$$

$$\Delta E = E_2 - E_1$$

$$\Delta E = \frac{-me^4}{8\epsilon_0^2 n_2^2 h^2} - \left(\frac{-me^4}{8\epsilon_0 n_1^2 h^2} \right)$$

$$\Delta E = \frac{me^4}{8\epsilon_0 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\Delta E = h\nu$$

$$= \frac{hc}{\lambda}$$

$$= hc \left(\frac{1}{\lambda} \right)$$

$$\Delta E = hc\bar{\nu}$$

$$\bar{\nu} = \frac{\Delta E}{hc}$$

$$hc\bar{\nu} = \frac{me^4}{8\epsilon_0 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\bar{\nu} = \frac{me^4}{8\epsilon_0 h^3 c} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$109736 \text{ cm}^{-1}$$

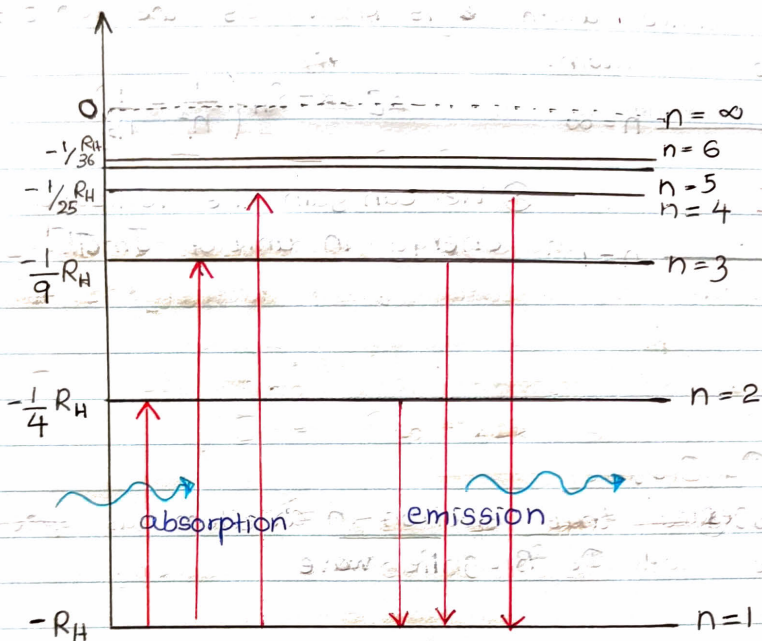
constant

$$R_H = \frac{(9.1 \times 10^{-31} \text{ kg}) \times (1.602 \times 10^{-19})}{8 \times (8.654 \times 10^{-12})^2 \text{ C}^2 \text{ s}^2 \text{ kg}^{-1} \text{ m}^3} \times 6.6 \times 10^{-34} \text{ Js}$$

$$R_H = 1.09736 \times 10^7 \text{ m}^{-1}$$

Day 04

23/01/2022



n = The Principle Quantum Number

$\uparrow / \downarrow$ = Electron transmission

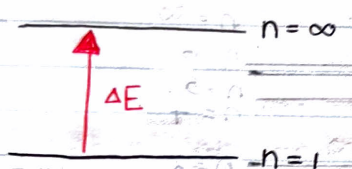
- We can successfully apply this Bohr theory to all H^+ and H^+ -like systems.
 - $\rightarrow \text{He}^+$
 - $\rightarrow \text{Li}^{2+}$

And we can't use The Bohr theory for the systems which have fewer more than one electron.

Calculation of Ionization Energy of the H atom



- (*) The amount of energy required to remove an electron from an isolated atom is known as the ionization energy of H atom.


$$\Delta E = -R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

(*) We can gain the ionization energy in unit of Jmol⁻¹!

WAVE PARTICLE DUALITY

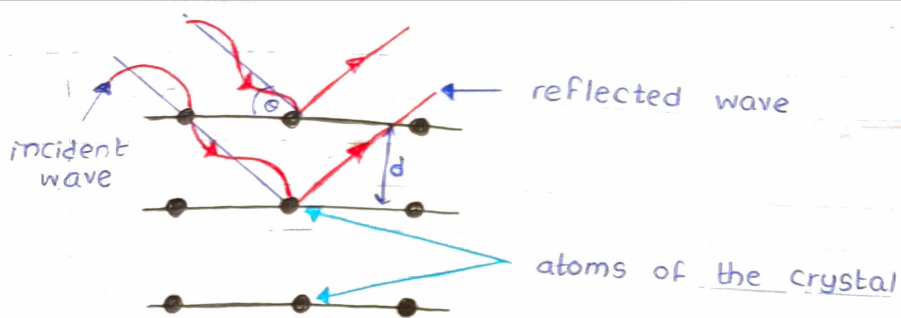
Louis De Broglie

- He suggested that electron in an atom has specific associated with De Broglie wave

- Evidences - wave nature of electrons

- ① Diffraction → *विचलन*
- ② Reflection → *परावर्तन*
- ③ Interference → *विवर्धन*

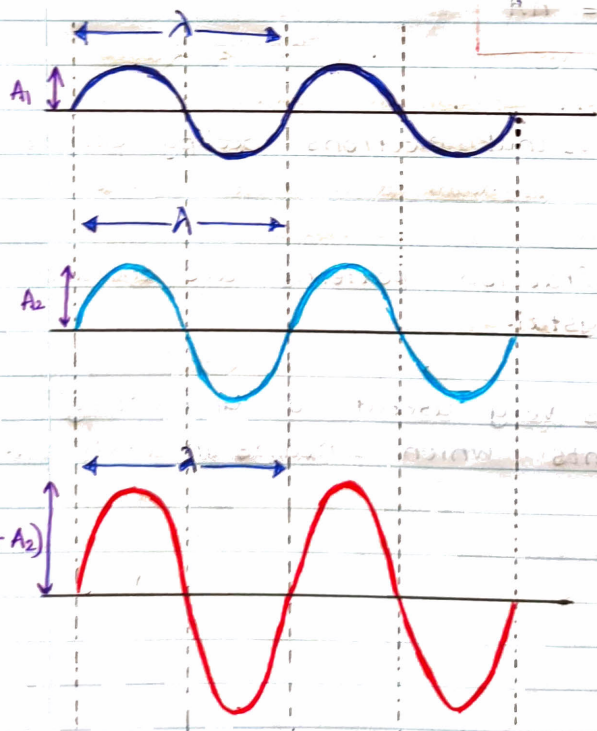
- Davison and Germer showed that a beam of electrons could be diffractive by the surface of an ionic crystal. (Similar to x-rays)



- The long second wave travels larger distance $\rightarrow (AB + BC)$

$$(AB + BC) = n\lambda \quad (\text{integral multiples of } \lambda)$$

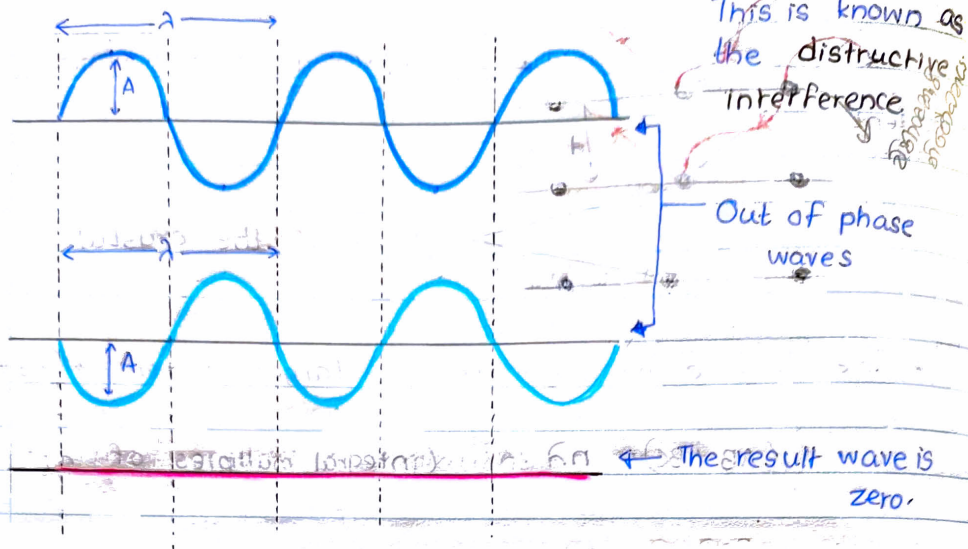
- If we mix two wave that have different amplitudes but with same wave length, the result will be a wave with a higher amplitude than the given two waves. Also the result wave's frequency is same as the two waves.



This incident is known as Constructive Interference.

The overlapping waves are in phase.

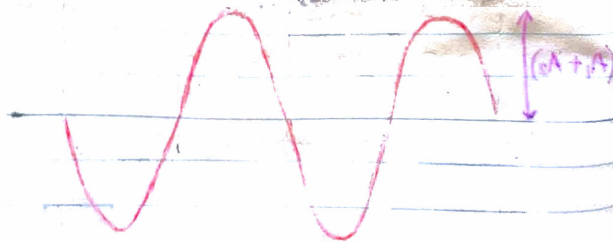
Constructive Interference



- This shows that in phase waves falling on the metal surface are interfered constructively. So this theory is known as The Bragg Law.

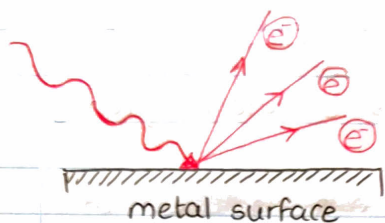
$$2d \sin \theta = n\lambda$$

- This diffraction shows that electrons occupy properties of waves.
- And also these diffraction patterns are characteristic for a particular crystal.
- So, this theory is very useful for identifying unknown elements which contain in a crystal.



- Another very important incident that we where we use the wave property of electrons is that, Electron Microscope.

PHOTOELECTRIC EFFECT

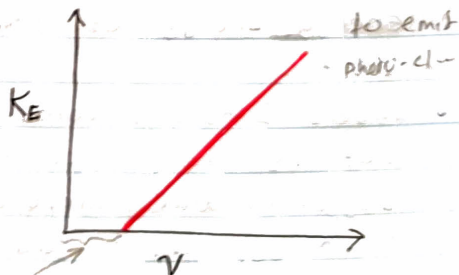
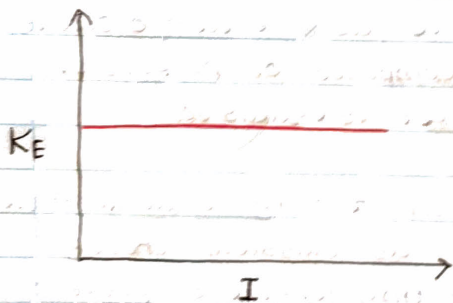


The photo-electron effect is the emission of electrons when electro-magnetic radiation, (such as light) hits a material. Electrons emitted in this matter are called photoelectrons.

The kinetic energy of the photoelectrons.

$$KE = \frac{1}{2}mv^2$$

- Classical mechanics wasn't able to explain this.
- The kinetic energy of photoelectrons is independent on the intensity of incident radiation.
- But the kinetic energy depends on the frequency.



This is zero because no electrons are ejected in this frequency range.

have to give the relevant amount of energy that bond of electron that occur with the needs to break the ^{retic} time

Conclusions

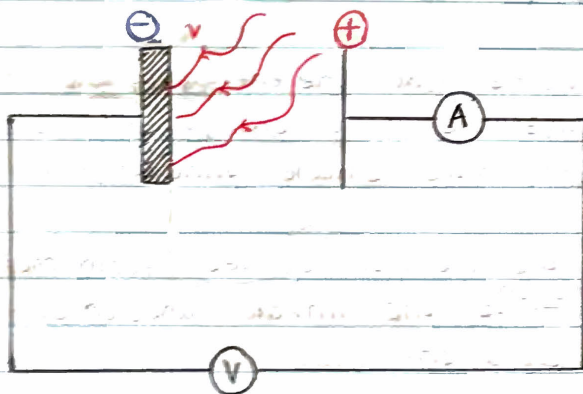
- ① Photoelectrons are generated instantly upon irradiation with appropriate frequency of radiation.
- ② Number of photoelectrons emitted depends on the intensity of incident of radiation.
- ③ Energy of the electrons depends on the frequency of the incident of radiation.
- ④ There's a maximum frequency of incident radiation called threshold frequency (ν_0) below which electrons are not ejected from the metal surface.

Day 05 → 29/01/2021

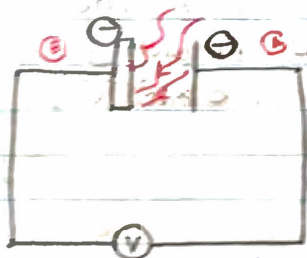
- So, the above ~~range~~ the range of the frequencies electrons are emitted from the metal surface.
- In Plank's Quantum Hypothesis of light, Plank explained that light is not continuous.
- The radiation can be considered as certain discrete amount of energy. Light is composed of packets of energy. Energy of one packet is equals to $h\nu$.
- Albert Einstein came up with a very crucial statement that light is not contribute as classical theory suggests, but composed of tiny discrete packets of energy, called quanta.
- G.N. Lewis gave this quanta a name → photon

- When the light strikes the metal surface electrons in atoms utilize the energy of impinging photons in two ways.
- When light beams falls on the metal surface, the part of the energy is used to overcome the attractive forces of electrons to the metal lattice, is known as Workfunction ϕ . Therefore eject an e^- from the metal surface.

$$h\nu = \phi + \frac{1}{2}mv^2$$



Energy of moving electron = ~~eV~~ $eV = \frac{1}{2}mv^2$



(A) is negatively charged. It repels electrons. When we increase the voltage there will be a moment that the motion of electrons stops. This particular high voltage is known as stopping potential.

$V = K.E. \text{ of electron}$ ~~eV~~ $eV_s = \frac{1}{2}mv^2$

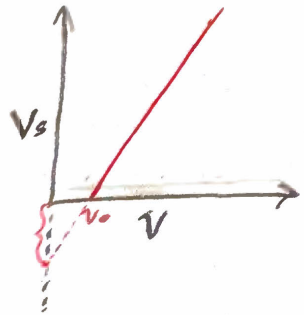
$$h\nu = \phi + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = h\nu - \phi$$

$$-eV_s = h\nu - h\nu_0$$

$$V_s = \left(\frac{-h}{e}\right)\nu + \frac{h\nu_0}{e}$$

$$\text{slope} = \left(-\frac{h}{e}\right)$$



- By all of this observations we can conclude that light has particle nature.
- Normally in the visible light the energy is low. But the UV rays have more energy than visible light because they have smaller wave lengths.
- The energy of the electron ejected from metal by UV ^{irradiation} ~~with radiation~~ is the linear function of the frequency of radiation.

④ A photoelectric surface has a workfunction of 4 eV. What is the maximum velocity of the photoelectrons emitted by light of frequency 10^{15} Hz? ($h = 6.6 \times 10^{-34}$ Js) ($e = 1.6 \times 10^{-19}$ C)
 ($m = 9 \times 10^{-31}$ kg)
 (1 eV = 1.602×10^{-19} J)

$$h\nu = \phi + \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2(h\nu - \phi)}{m}}$$

$$v = \sqrt{\frac{2[(6.6 \times 10^{-34} \text{ Js} \times 10^{15} \text{ Hz}) - (4 \times 1.602 \times 10^{-19} \text{ J})]}{9 \times 10^{-31} \text{ kg}}}$$

$$V = \sqrt{\frac{2(6.6 \times 10^{-19} - 6.408 \times 10^{-19}) \text{ J}}{9 \times 10^{-31} \text{ kg}}}$$

$$V = \sqrt{\frac{2 \times 192 \times 10^{-22}}{9 \times 10^{-31}}}$$

$$V = \sqrt{\frac{2 \times 192 \times 10^9}{9}}$$

$$V = 2.06 \times 10^5 \rightarrow \underline{\underline{V \approx 2.1 \times 10^5 \text{ m/s}}}$$

- Broglie suggested that particles like electrons have also a wave nature.
- Let's see how the momentum of a particle relates to the wave length is that

$$\lambda = \frac{h}{p}$$

$$\lambda = \frac{h}{mv}$$

Broglie wave length

$m = 157 \text{ g}$  $V = 42 \text{ m/s}$
 $\textcircled{e}$ $3 \times 10^6 \text{ m/s}$
 $m = 9.8 \times 10^{-31} \text{ kg}$

$$\lambda = \frac{6.6 \times 10^{-34} \text{ Js}}{9.8 \times 10^{-31} \times 3 \times 10^6 \text{ m/s}}$$

$$\lambda = \frac{6.6}{9.8 \times 3} \times 10^{-9}$$

$$\underline{\underline{\lambda = 2.4 \times 10^{-10} \text{ m}}}$$

For the ball

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{6.6 \times 10^{-34} \text{ Js}}{157 \times 10^3 \text{ kg} \times 42 \text{ m/s}}$$

$$\lambda = \frac{6.6}{157 \times 42} \times 10^{-31}$$

$$\underline{\underline{\lambda = 1 \times 10^{-34} \text{ m}}}$$

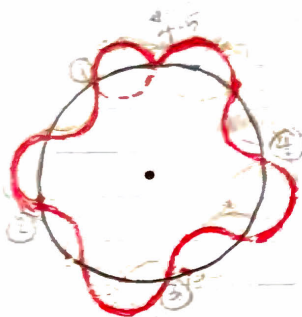
• The wave length of the cricket ball is so small compared to the electron. So, we can assume that cricket ball nearly doesn't show any wave properties.

Its partical nature is determinant.

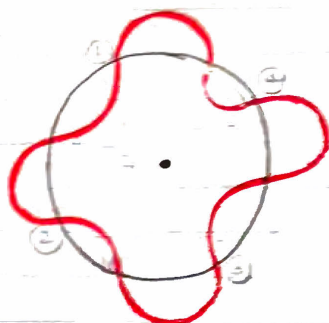
- The wave length of the electron is very larger compared to the radius of the electron. The λ is incomparable and detected. ~~For~~ Therefore we can say that wave-nature is exhibited in moving electrons.
- Let's try to apply this wave nature to the electron or to a Hydrogen atom.

The association of the wave-nature of electron is ---

BOHR MODEL OF THE ATOM



Not Constructively
Fitting to the Bohr
Orbit



Constructively fitting to
the Bohr Orbit
 $n=4$

- So we can say that electron exists ^{as a} Standing wave in each orbit.
- In order to fit a wave length electrons can be only in those orbits
- To Bohr atom the length of the orbit ($2\pi r$) ^{should} ~~to be~~ equal to a whole number multiple of the wave length.

$$n\lambda = 2\pi r$$

$$\frac{nh}{mv} = 2\pi r$$

$$r = \frac{nh}{2\pi mv}$$

Bohr Radius

- No. _____ Date: ____/____/____
- This equation is ^{an} agreement with the Bohr's quantization hypothesis.

HEISENBERG'S UNCERTAINTY PRINCIPLE



A planet



- We can accurately locate exact position and the momentum ~~a~~ of the planet at the same time.
- This principle says the exact position and the momentum of a particle (electron) cannot be determined simultaneously with the desired accuracy.

Δx = uncertainty in the position

Δp = uncertainty in the momentum

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

- If one quantity is measured accurately, the other quantity becomes less accurate.
- Now let's apply Heisenberg's uncertainty principle theory to explain the behaviour of microscopic objects such as, electrons, protons and ~~new~~ neutron.

- Q An electron is moving in 600 m s^{-1} velocity. So, its accuracy is 0.005% . Calculate the certainty with which we can locate the position of the electron.

$$\Delta x = ?$$

uncertainty of momentum

$$\text{is } \Rightarrow \Delta p = \left(\frac{0.005}{100} \right) mv$$

$$\Delta p = \frac{0.005}{100} \times 9 \times 10^{-31} \text{ kg} \times 600 \text{ m s}^{-1}$$

$$\Delta p = 5 \times 10^{-3} \times 10^{-2} \times 9 \times 10^{-31} \times 6 \times 10^2$$

$$\Delta p = 2.7 \times 10^{-28}$$

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \geq \frac{6.6 \times 10^{-34} \text{ Js} \times 7}{2.7 \times 10^{-28} \times 4 \times 22}$$

$$\Delta x \geq \frac{6.6 \times 7}{2.7 \times 4 \times 22} \times \frac{1}{10^6}$$

$$\Delta x \geq 0.00192 \text{ m}$$

$$\Delta x \geq 1.9 \text{ mm}$$

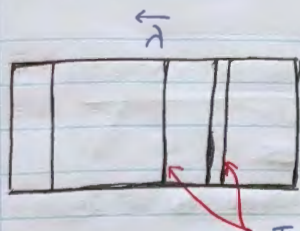
30/01/2022
Day - 06

- The relationship between E and n is,

$$E \propto \frac{1}{n^2}$$

$n \rightarrow$ principle quantum number.

This means that the concept of an electron as a tiny particle is involved to do good.

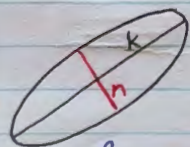


- The lines we can gain in the emission spectrum due to the electron transmission between two energy levels.
- But ~~these~~ lines when these each line was checked under a high resolution spectrometer, those lines ~~are~~ are consisted of a collection of another lines.
- So that means the energy levels or the orbits are consist of some other sub-energy levels.
- Scientist called Sommerfeld explained that existence of closely spaced ~~in~~ placed lines in terms of elliptical orbits. (What was Bohr explained that they were circular orbits.)
- He named these elliptical ~~qu~~ orbits as another quantum number. This was named as l (azimuthal quantum number).

$$n = 1, 2, 3, 4, 5 \dots \infty$$

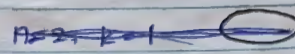
$$l = 0, 1, 2, 3, 4 \dots \infty - 1 \quad (n-1)$$

- In an elliptical orbit two diameters can be identified:



$\frac{n}{k}$ describes the shape of the orbit.

for $n=1, k=1$



$n=2, k=1$



$$n=3, k=3, 2$$



• Fundamental equation of quantum mechanics

► This was introduced by Erwin Schrödinger

$$\overset{\text{imaginary number}}{i} \frac{h}{2\pi} \frac{\partial \psi}{\partial t} = \hat{H} \psi$$

(P.S.) wave function

Hamiltonian operator

Partial derivative with respect to time

$$v = \frac{g}{t}$$

$$i \frac{h}{2\pi} \frac{\partial \psi}{\partial t} = \hat{H} \psi$$

time dependent Schrödinger equation.

$$\hat{H} \psi = E \psi$$

time independent

• we can apply this Schrödinger equation to the Hydrogen atom

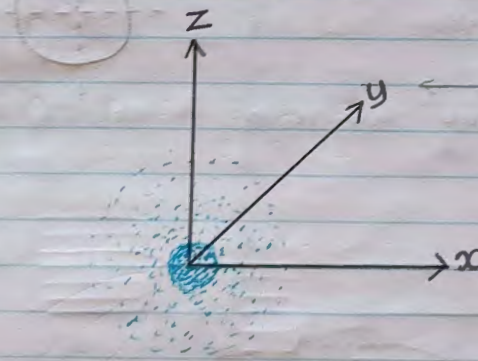
$$\psi_1 \rightarrow E_1$$

$$\psi_2 \rightarrow E_2$$

doesn't have physical meaning and interpretation



- Max Born was the person who gave a physical meaning for this sign. What he proposed that the absolute value of ψ^2 in a small space interval of space has a meaning. Physical interpretation represents the probability of finding the system in a small space.
- The solution of the Schrödinger equation for the Hydrogen atom resulted a series of mathematical functions called wave functions.
- ψ^2 gives an idea for the location of the electron in an atom.
- According to quantum mechanics electron can move only in a region of space not and not in a fixed orbit.
- Also, we can define ψ^2 in another different way.
It is the probability of finding an electron in a particular region of space.

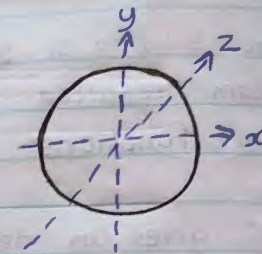
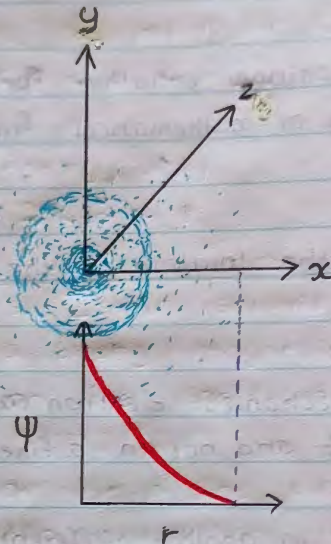


- The probability of region of space where the probability density is fairly high. This space where the probability of electron can exist is known as the orbit.

LOWEST ENERGY LEVEL OF HYDROGEN ATOM

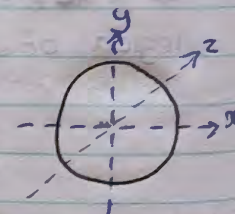
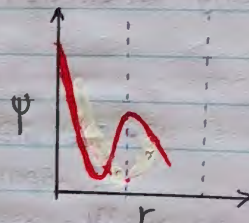
- The lowest energy level of (H) is **1s**.
- The next (second) excited energy level is of (H) atom is **2s**.

1s



- The electron occurred by the space is spread in all directions

2s

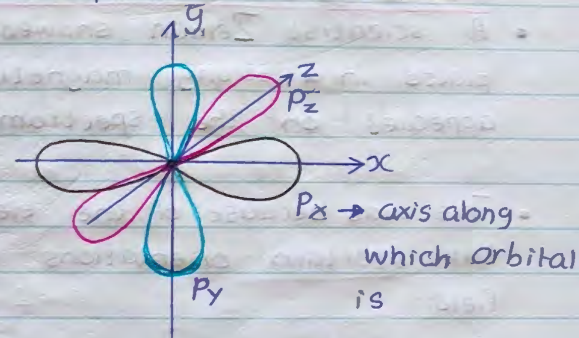
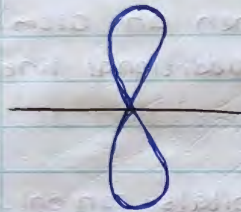


Azimuthal Quantum Number (l)

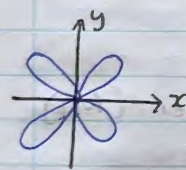
- $n=1$ $l=0$
- $n=2$ $l=0, 1$
- $n=3$ $l=0, 1, 2$
- $n=4$ $l=0, 1, 2, 3$
- $n=5$ $l=0, 1, 2, 3, 4$
- $\vdots$
- $n=n$ $l=0, 1, \dots, (n-1)$

- This describes the angular momentum of the electron, relative to the nucleus.

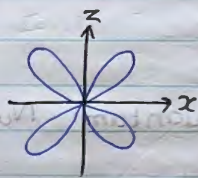
- This also determines the shape of the orbit.



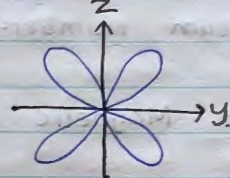
d_{xy}



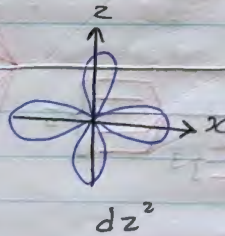
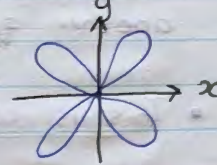
d_{xz}



d_{yz}



$d_{x^2-y^2}$



- There's an equal probability of being an electron in either lobe p orbital.

Ma

Day 07

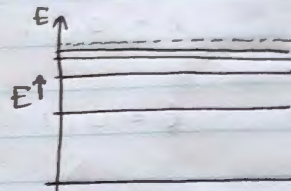
- $n=3$ energy level has 9 orbitals
- $n=4$ energy level has 16 orbitals

- If we look at the order of the energy of sub-energy levels for a specific principal quantum number it increases s orbitals to f orbitals.

$$s < p < d < f$$

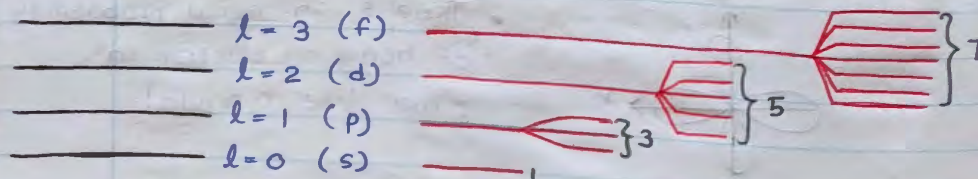
principal energy level

$$1 < 2 < 3 < 4$$



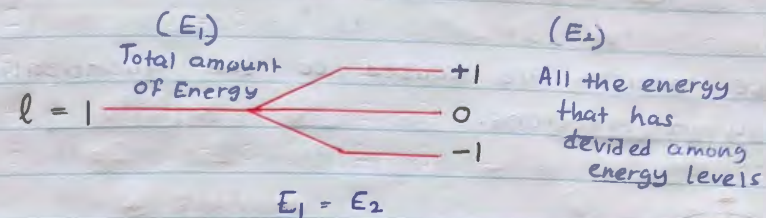
Magnetic Quantum Number (M_l)

- A scientist Zeman showed that when an atom is placed in a strong magnetic field additional lines appeared on the spectrum. (emission?)
- This was because of the electrical orbitals can only take certain orientations with respect to the magnetic field.
- So, each of these orientations is associated with another quantum number.
- This is called Magnetic Quantum Number (M_l)
This was the third.

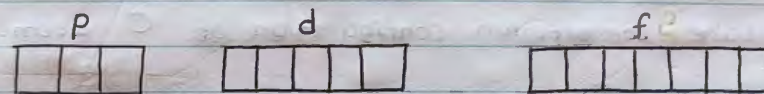


- An Azimuthal quantum is split in to $(2l+1)$ number of magnetic quantum numbers.

- m_l can be any Azimuthal Quantum number such as $+l \dots 0 \dots -l$.



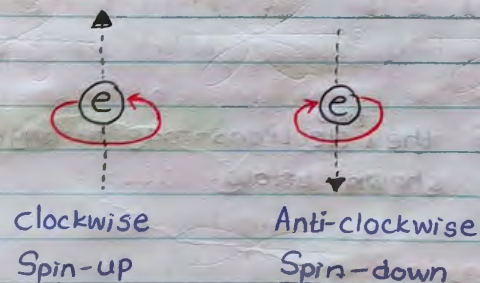
If the amount of energy is equal in two energy levels, they ^{are} known as degenerate energy levels.



Spin Quantum Number (M_s)

* Electron Spin

- The e^- behaves if it was spinning about an axis through its centre.



- The two directions of spin corresponds to the possible values for the Spin Quantum number (M_s).

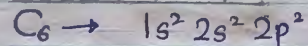
- So it is obvious that e^- has two motions
i.e. ① Spin motion
② Orbital motion

- These motions are caused to the quantization of energy in electrons.

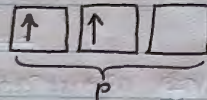


$$M_s = +\frac{1}{2} - \frac{1}{2}$$

- Let's take the electron configuration of C atom.



Ex:- 2p electrons



$$n=2$$

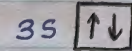
$$l=1$$

$$m = \pm 1$$

$$m_s = +\frac{1}{2}$$

3s electrons

We can Let's take if the ^{two} electrons may arrange in two ways as shown below.



Arrangement 1

$$n=3 \quad n=3$$

$$l=0 \quad l=0$$

$$m=1 \quad m=1$$

$$m_s = +\frac{1}{2} \quad m_s = -\frac{1}{2}$$



Arrangement 2

$$n=3 \quad n=3$$

$$l=0 \quad l=0$$

$$m=1 \quad m=1$$

$$m_s = +\frac{1}{2} \quad m_s = +\frac{1}{2}$$

Pauli Exclusion Principle

⊛ This principle says that no two electrons in an atom can have the same set of 4 quantum numbers.

⊛ In other words

1) No more than two electrons can occupy the same orbital

&

2) Two electrons in the same orbital must have opposite spins.



✗

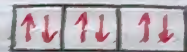


✗



✓

- Therefore, it is obvious that the arrangement 2 of electrons can be existed.
- We say an orbital is stable when it is fully filled or halfly filled.



fully filled



half filled.

Hund's Law

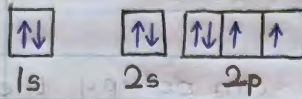
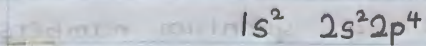
Every orbital in a subshell is singly occupied with one electron before any one orbital is doubly occupied, and all electrons in singly occupied orbitals have the same spin.

(In other words, In the orbitals of the same subshell, electrons are filled singly first before pairing starts.)

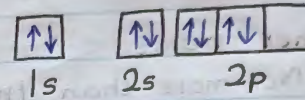
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Example -:

Let's take the electron contribution of O atom.



correct (✓)



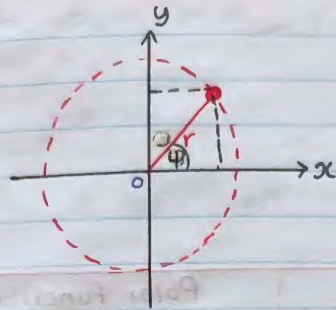
incorrect (✗)

Shrodinger

Day 18 - 12/02/2022

- Let's see how we can use simple mathematics to understand the shapes of orbitals.
- We realized that the Shrodinger equation the mathematical expression and the solution for the Hydrogen atom are mathematical expressions.
- ~~We can~~
- We can represent this wave function for the solution for the Shrodinger equation to understand in spherical polar co-ordinates of the electron in (H) atom.
- Now, the solution for the wavefunction for the solutions of the Shrodinger equation in the (H) atom signify the orbitals of the (H) atom.

- If we consider the motion of a particle which moves in a constant speed in a particular plane



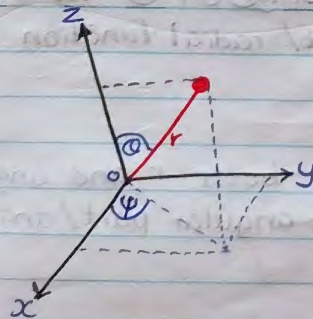
- This particle is moving r distance away from the nucleus.

- Let's say that the linear distance (r) makes an angle which has a magnitude of ψ with the x axis.
- This represents the circular motion of a particle in xy plane.
- In terms of Cartesian co-ordinates (x, y)

$$x = r \cos \psi$$

$$y = r \sin \psi$$

- If the motion of the particle is not confined to the xy plane we have to use spherical-polar co-ordinates.



- r is the distance from the position of the particle from the origin.
- θ is the angle which the projection of vector r makes with respect to the z axis.
- ψ is the angle which the projection of vector r makes with respect to the x axis.

- In terms of spherical-polar co-ordinates and the angles θ and ψ .

$$x = r \sin\theta \cos\psi$$

$$y = r \sin\theta \sin\psi$$

$$z = r \cos\theta$$

Orbital	Quantum Numbers			Polar function
	n	l	m	Radial wavefunction
1s	1	0	0	e^{-r}
2s	2	0	0	$(2-r)e^{-r/2}$
2p _x	2	1	0	$re^{-r/2} \cos\theta$
2p _y	2	1	1	$re^{-r/2} \sin\theta \cos\psi$
2p _z	2	1	-1	$re^{-r/2} \sin\theta \sin\psi$

- If the polar function depends only on r we refer that as a radial part/radial function of the orbital.
- If the polar function depends both r and angles θ and ψ it is known as an angular part/angular function of the orbital.

$\psi \rightarrow$ The probability of finding the electron in space of the atom.

- Each orbital is a product of a part which depends on the distance 'r' and the other which depends on the angles θ and ϕ .

$$\psi^2 dx dy dz$$

This is the direct measure of the probability of finding the electron in the volume $dx dy dz$ (dv) with an Energy E.

We always get the lowest energy of an atom, when the atom is in the ground stage.

- So, the ^{solution of the} Schrödinger equation to the (H) atom / to the lowest value of energy of the electron



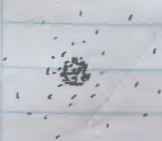
$$\psi = e^{-kr}$$

$k \rightarrow$ constant

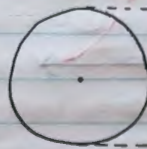
So, we can see the solution take the type of exponential for (powered to the e)

When the distance / r increases the ψ decreases

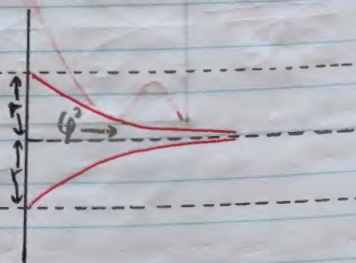
The highest value of ψ can be obtained when the $r=0$.



Charge cloud representation



The boundary surface



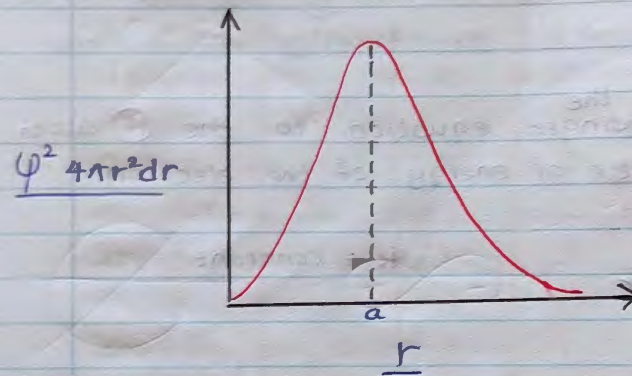
plot of ψ^2 vs. r

The radial probability of 1s orbital

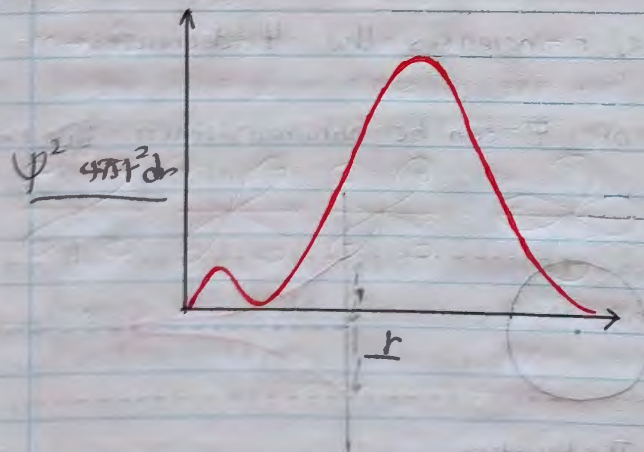
The volume $\rightarrow 4\pi r^2 dr$

radial probability $\rightarrow \psi^2 4\pi r^2 dr$

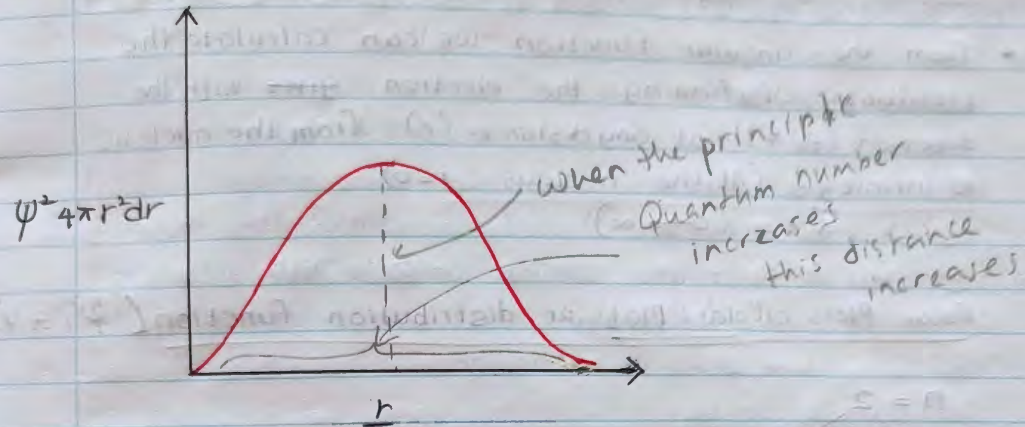
$\psi^2 \downarrow$ $r \uparrow$
 $V \uparrow$ $r \uparrow$



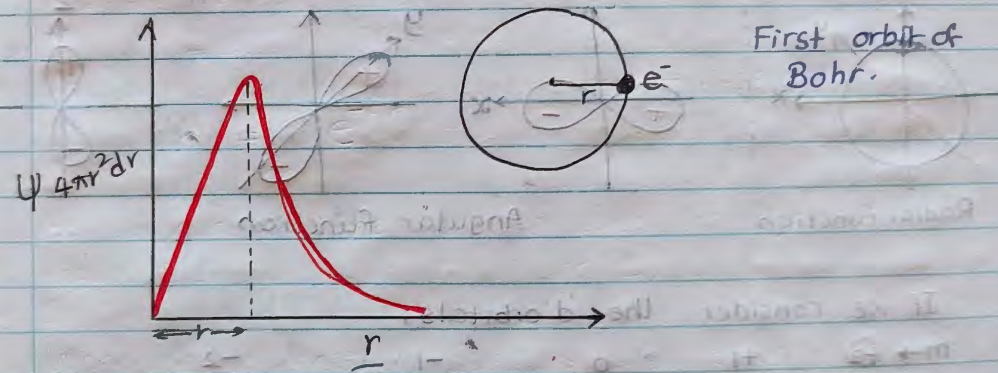
2s orbital



2p orbital



H atom



Angular Function

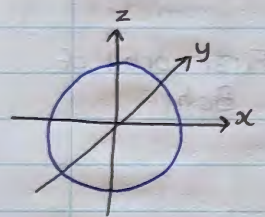
- From the angular function we can calculate the probability of finding the electron gives with the direction (θ, ϕ) at any distance (r) from the nucleus to infinity. (At the nucleus $r=0$)
($r = (0 \rightarrow \infty)$)

Polar Plots of the angular distribution function $[\psi^2(\theta, \phi)]$

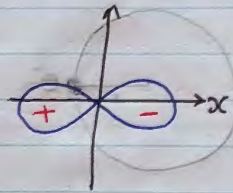
$n = 2$

S

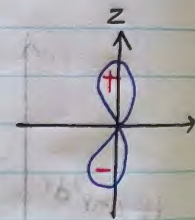
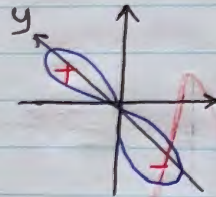
P



Radial function



Angular function



If we consider the d-orbitals,

$m \rightarrow +2$

$+1$

0

-1

-2

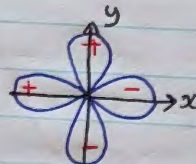
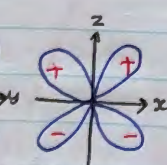
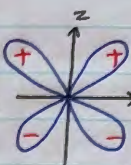
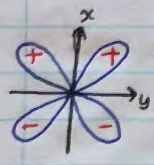
d_{xy}

d_{yz}

d_{xz}

$d_{x^2-y^2}$

d_{z^2}



orbitals lies between the directions.

orbitals lies along the directions

- Bonding is a result of overlapping of electron clouds, over the wave functions. Let's explain this in a simple way.

Bonding is described for a covalent bond ~~the~~ is the changes ~~the~~ the electrons between the atoms. For an example, let's take the simple Hydrogen (H_2) molecule it forms by two (H) atoms. So the two electrons - one electron comes from one (H) atom and the other electron comes from the other (H) atom. So, the two electrons are shared with the two nuclei.

Day 09 - 13/02/2022

$$4\pi r^2 \psi^2 dr$$

$$r = 0$$

$$r = \infty$$

total probability of finding the electron in all space must be unity.

$$\int_0^{\infty} 4\pi r^2 \psi^2 dr = 1$$

$$\int_0^{\infty} \psi^2 r^2 = 1$$

volume element

This is called as the normalization condition

Valence Bond Theory

This theory assumes that all bonds are localized bonds formed between two atoms by the donation of an electron from each atom. That is how a covalent bond is formed.

STRUCTURE AND BONDING

Covalent Bonds

Properties of Covalent Compounds

- ① Usually have low melting points and low boiling points.
- ② Most of them not soluble in water; they dissolve in organic solvents.
- ③ They ~~are~~ tend to be soft and relatively flexible.
- ④ Usually have lower enthalpies of fusion and vapourization than ionic compounds.
- ⑤ They tend to be more flammable than ionic compounds.
- ⑥ When some compounds in water they don't conduct electricity.
- ⑦ They don't exist in single molecules most of the time.

When we are learning about the covalent bonds there are two theories that are very helpful i.e;

- ① Valence Bond Theory
- ② Molecular Orbital Theory

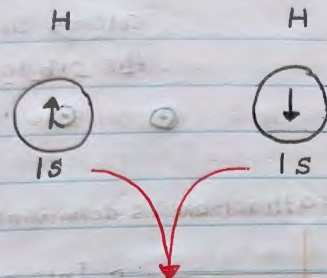
* These two ~~are~~ theories ~~at~~ were developed and applied by Linus Carl Pauling (1901-1994)
[Electron negativity in Organic Chemistry]

Valence Bond Theory

This theory assumes that all bonds are localized bonds formed between two atoms by the donation of an electron from each atom. That is how a covalent bond is formed.

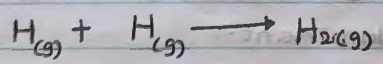
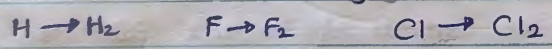
- This theory simply explained that the electrons can be shared when half filled orbitals in one atom overlaps in phase with half filled orbital in the other atom, while forming a covalent bond.

ex-:



we can say that the two 1s orbitals overlaps/mix to form a covalent bond to make H_2 molecule.

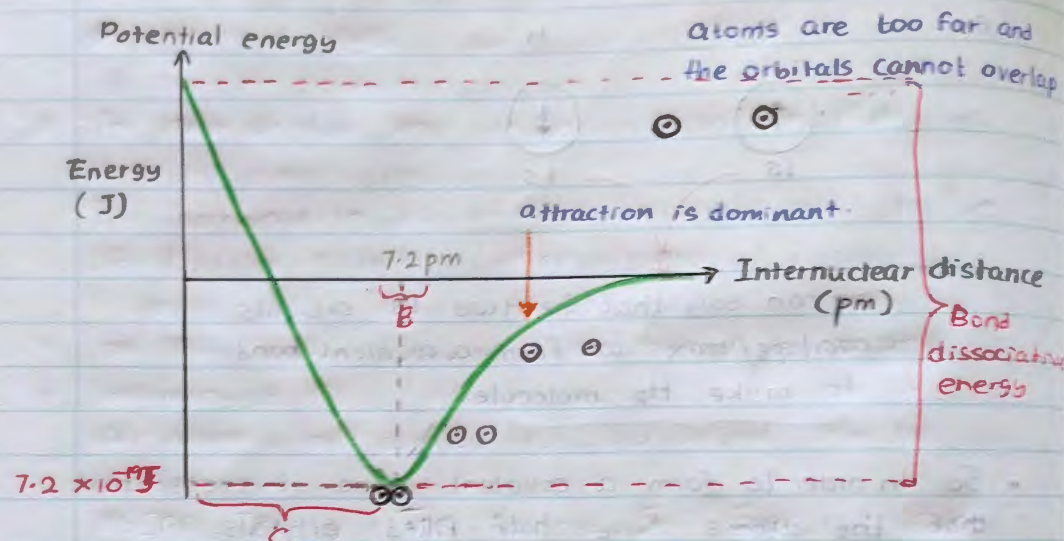
- So, in order to form a covalent bond, it is essential that the atoms have half-filled orbitals.
- Why H , F and Cl elements exist in nature as diatomic molecules instead of free atoms?



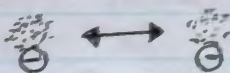
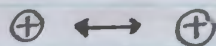
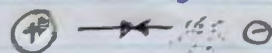
► $H_{2(g)}$ has lower energy than free $H_{(g)}$ atoms and also $H_{2(g)}$ is more stable.

Lewis Theory

- * This theory, the formation of bonds is explained by atoms transfer electrons with others.



- * When the atoms are too far the energy is close to zero that means comparatively to the H atoms there has no energy changed.
- * When we bring the H atoms closer and closer, the attraction is dominant.
- * What is attractionary repulsion
If we have two H atoms positively charged nucleus can attract negatively charged electron cloud.



- * When the atoms are too close repulsion is dominant.

No. _____ Date: ____/____/____

* ~~When~~ When the two (H) atoms comes closer their orbitals get overlaps and mixs. And the bond is formed.

* When the bond begins to form energy decreases.

* In the region B:

▶ It is the energy minimum point.

▶ This minimum amount of energy has calculated as $7.2 \times 10^{-19} J$.

▶ Here also repulsive forces are dominative.

* In the region C,

▶ the repulsive forces are dominative.

▶ The energy suddenly increases (after the minimum value).

* In the optimum distance the energy is minimum.

This is called as the bond length.

* If the repulsive forces are higher than the attraction, the bond cannot be formed.

If we try to do that the H_2 molecule will disssociate again.

* When the bond is formed (by the combination) of two (H) atoms energy is released.

* So, the H_2 molecule consists of lower energy compared to the two (H) atoms.

* Also we have to supply energy to break this bond between the two (H) atoms.

* When atoms are bound together in molecules the wave function ^{are} combined to produce new mathematical description

* The wave functions are combined by the linear combination of atomic orbitals.

* The new orbital that resulted are called Hybrid orbitals.

The combination of atomic orbitals to produce Hybrid orbitals are called Hybridization

* So the atomic orbitals can mixed/overlap and according to the linear combination of orbitals can result hybridized orbitals. / Hybrid Orbitals.

VSEPR Theory (The Valence Shell Electron Pair Repulsion Theory)

- This predicts the shape of Hybrid Orbitals.
- This theory is a model used to predict 3D molecular geometry based on the number of valence shell electron bond pairs among the atoms in a molecule or ion.

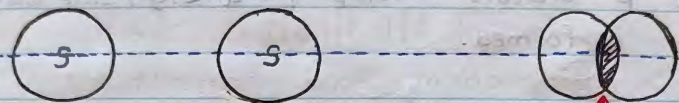
Hybridization

- There are several ways of overlapping orbitals.

① Linear overlap Along the intermediate axis.

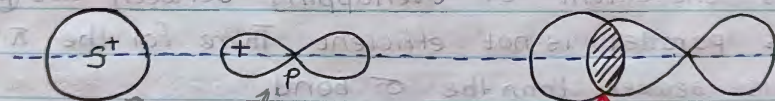
For an example let's take the two atomic orbitals of (H) atoms forming a H—H bond.

- ▶ 1s orbitals of (H) atoms.



The overlap takes place ~~on the~~ along the intermediate nucleus axis.

- ▶ s orbital and p orbital



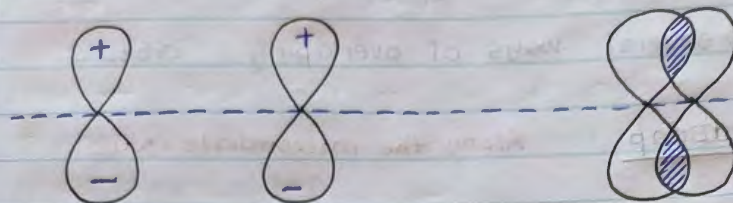
The wave function of sign of these should be same

These two should be in-face

- ▶ two p orbitals



② Sideway Overlap



- ▶ The electron density is on the opposite sides of the intermediate axis.
- ▶ If two p orbitals overlap parallelly / side ways π bond is formed.
- ▶ The strength of a covalent bond (π or σ) depends on the extent of overlap of the atomic orbitals involved.
- ▶ σ -bond is stronger than the π -bond. The reason for this is the extent of overlapping between two p orbitals parallel is not efficient. There for the π bond is weaker than the σ bond.
- ▶ When forming a covalent bond electron sharing takes place with the electrons in the outermost shell.



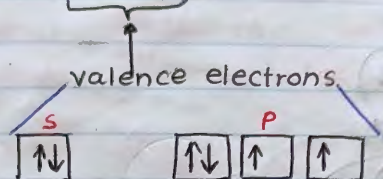
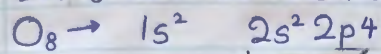
Practice problems

① H_2O



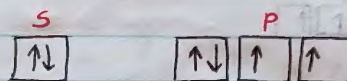
- There are six electrons in the valence shell of the Oxygen atom.

Let's take the electron contribution of a Oxygen:



This is the electron configuration of the oxygen atom in ground stage.

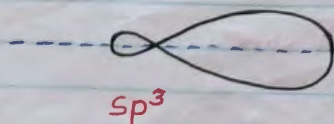
The Oxygen atom can form two covalent bonds with H atom.



↑
 sp^3 hybridization

This indicates that these sp^3 hybrid orbitals have 25% of s orbital characteristics and 75% of p orbital characteristics.

More characteristic of p means that p orbitals are much more diffused/extended away from the nucleus. s orbitals are closer to the nucleus.

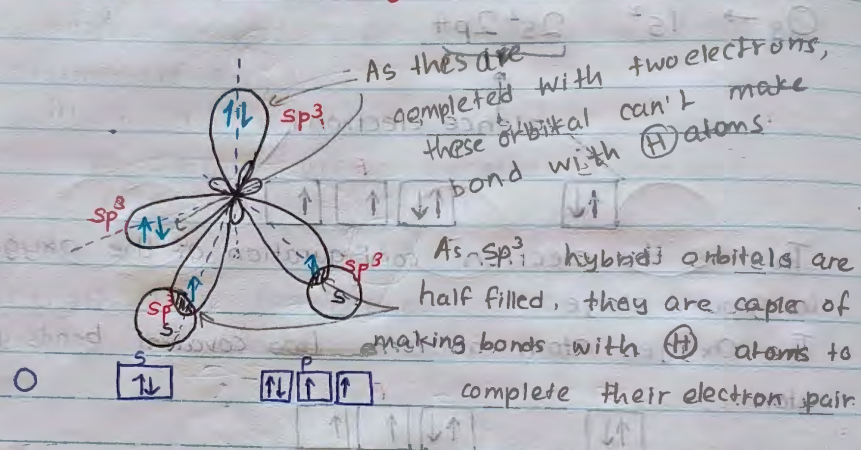


4 sp^3 hybridized orbitals are formed in a water (H_2O) molecule around Oxygen.

* Number of Hybrid orbitals formed is equal to the number of atomic orbitals mixed.

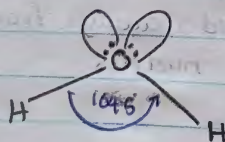
• How ~~is~~ these 3 forms of sp^3 hybrid orbitals orientate in space?

We have **Tetrahedral Arrangement** of H_2O in space.

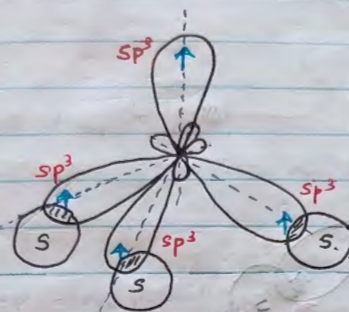
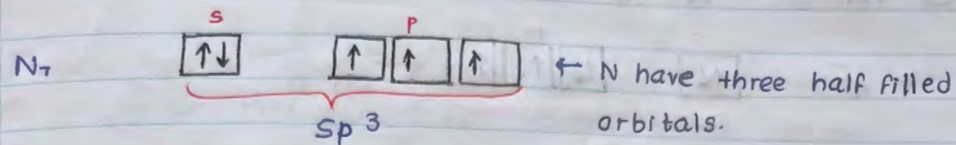
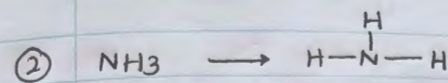


To determine the shape of the H_2O molecule, we can consider about the VSEPR electron pairs.

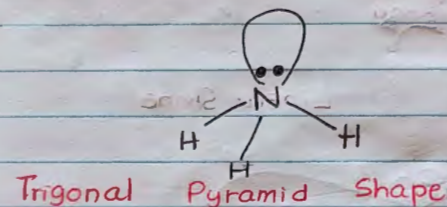
Bond pair - Bond pair repulsion < Lone pair - Bond pair repulsion < Lone pair - Lone pair repulsion



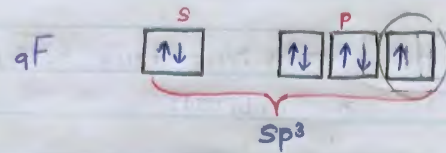
Bent/V shape



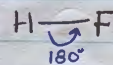
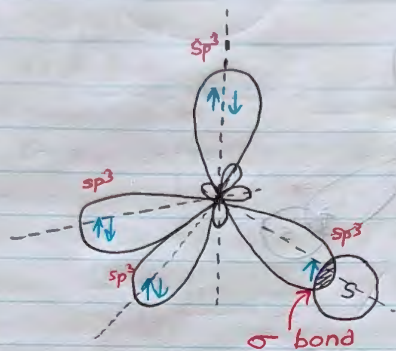
As every sp^3 orbital half filled, each one can make covalent bonds with $(1+)$ atoms.



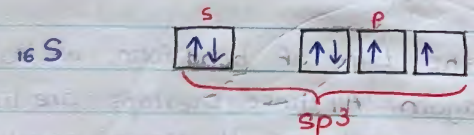
- We we consider about the type of bond form, we have look at the overlapping again. All these overlaps are linear overlaps. The sp^3 and s orbitals of H atoms overlap linearly along the longitudinal axis.
- So all the all the bonds between N and H (N—H) are σ bonds.
- So the shape of the Ammonia is Trigonal Pyramidal.



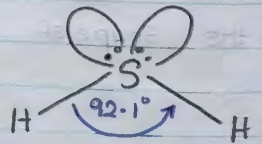
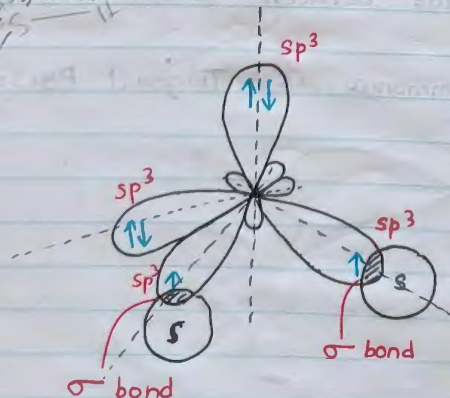
Only one half filled orbital are left. So only one (H) atom can form a covalent bond with (F) atom



Linear Shape

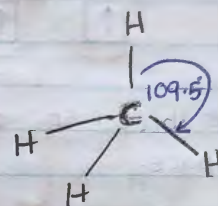
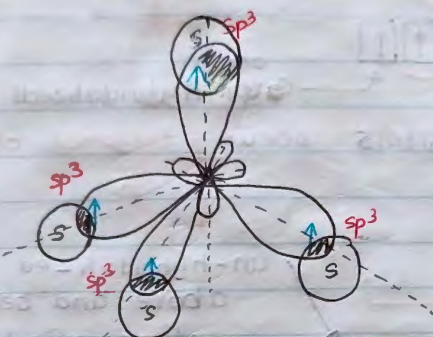
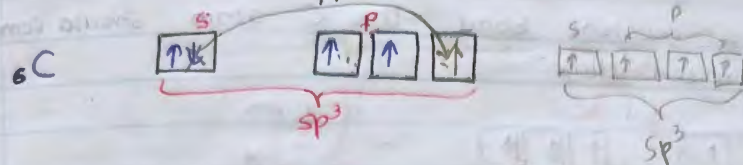
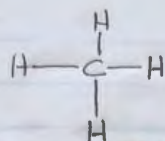


Two half filled orbitals are left. So only two (H) atoms can make covalent bonds with (S) atoms.



Bent (Angular)

⑤ CH_4

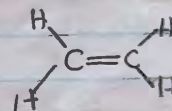


Tetrahedral shape.

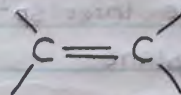
Day 10

Let's consider about further more about hybridization of orbitals.

Let's take C_2H_4 as an example



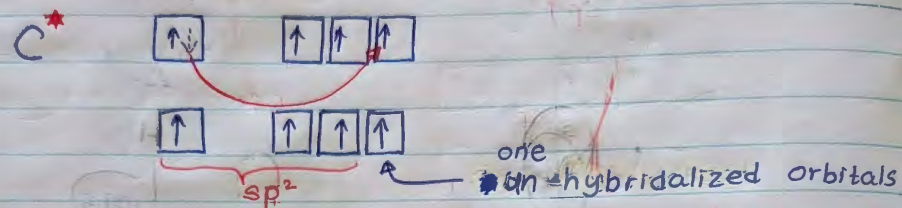
The contribution of electrons of the valence shell in the C atoms at the ground stage is $1s^2 2s^2 2p^2$



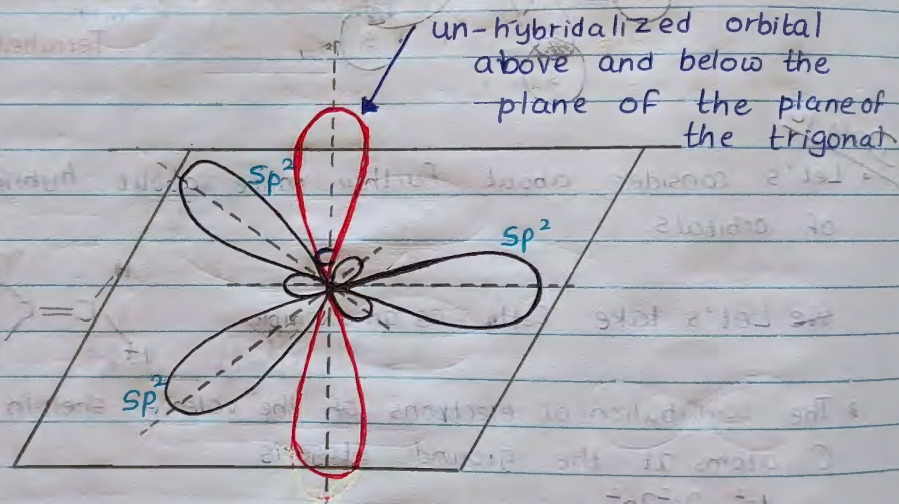
- In order to form a covalent bond, there should be 3 half filled orbital
- But this C atom can form only 2 bonds. As the sketch above shows there are three bonds around the one Carbon atom, there should be some way to make three

bonds

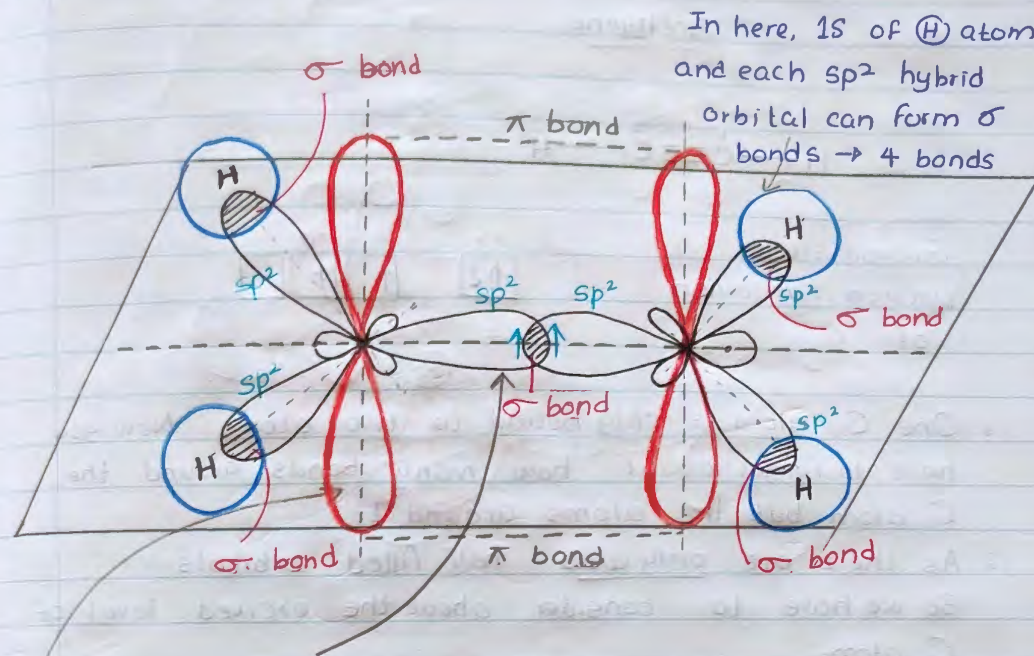
- To form a covalent bond the C atom should come to an excited stage.



- these three sp^2 orbitals occur trigonal plane shape in space.



Always in sp^2 we have three sp^2 orbitals which are arranged in a trigonal plane.



These two electrons can form a σ bond which have a higher density / higher possibility of electron, ~~are~~ cylindrically symmetrical around the intermediate axis.

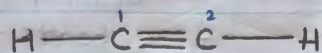
The two unhybridized orbitals there are two single electrons. These orbitals are parallel to each other. So they can overlap sideways to form a π bond, they overlap sideways. This π bond is cylindrically symmetrically located around the intermediate axis.

A π bond is weaker compared to the σ bond because sideways overlapping of unhybridized orbitals to form π bonds is not efficient. The orbital overlap along the intermediate ^(it is maximum) is efficient, so that the σ bond makes much stronger.

Conclusion

If we have a C atom with four bonds but 3 atoms bound around it, the C has sp^2 hybridization.

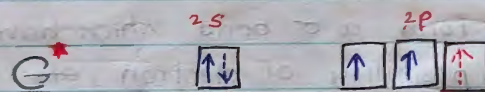
Acetylene



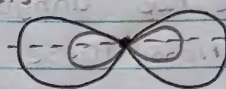
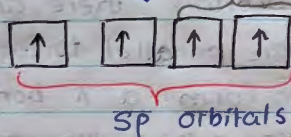
Ground stage
electron contribution
of C



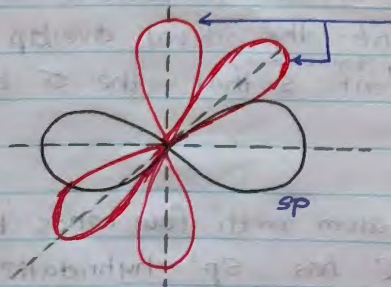
- One C atom is only bound to two atoms. Now you have to forget about how many bonds around the C atom but the atoms around it.
- As there are only two half filled orbitals. So we have to consider about the excited level of C atom.



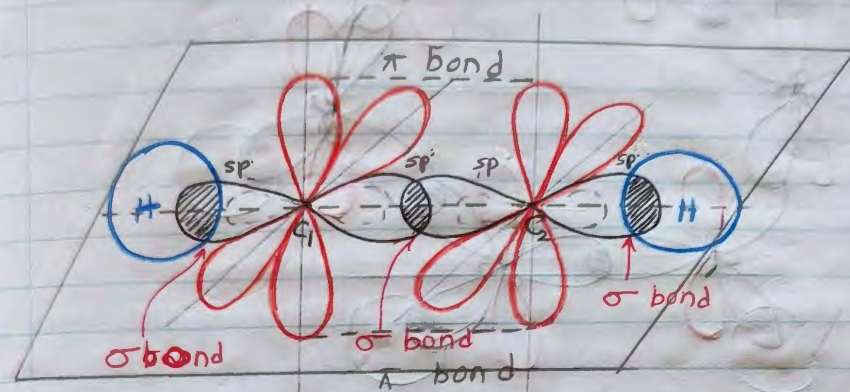
Two un-hybridized orbitals.



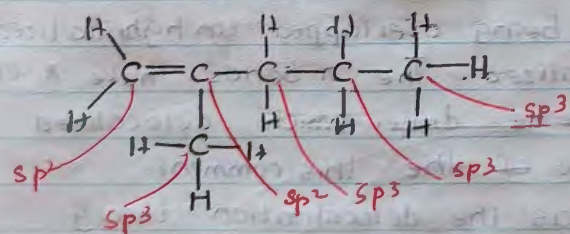
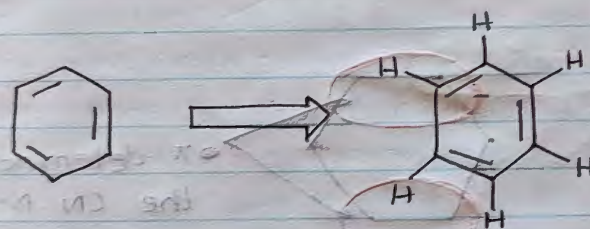
Now let's have a look at the 3D orientation of the orbitals of this molecule.



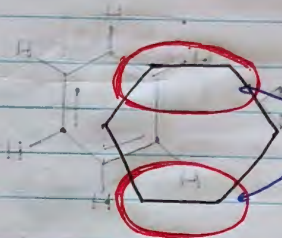
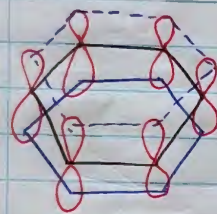
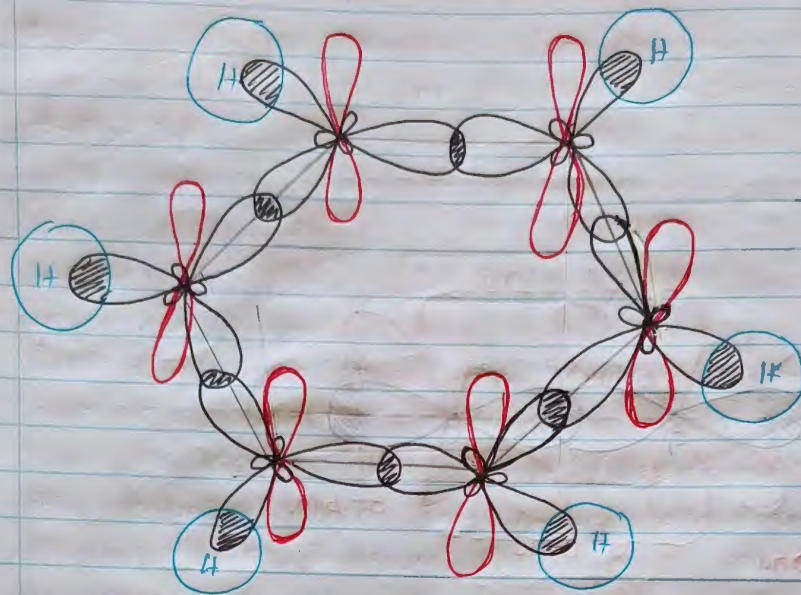
Two-unhybridized orbitals.



Benzene



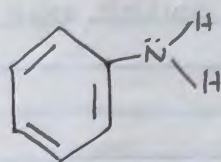
The shape is given / or the orientation is given by these functions.



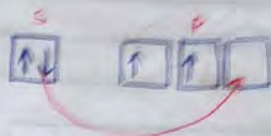
• π determine
the CN Networking

- The possibility of being overlapped un hybridized electrons to delocalized. The π bonds make π clouds which are draw to be draw, which delocalized above and the below of the this comment. This is known as the delocalization theory.

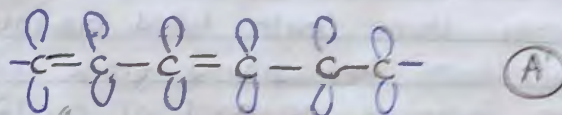
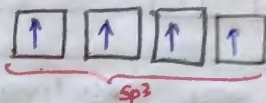
~Alinine~



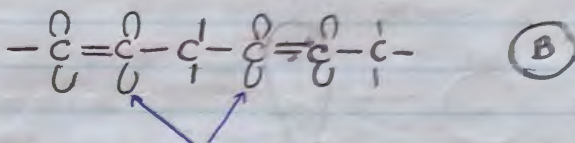
Ground
stage N
7N



Excited way
G*

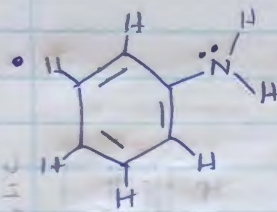


- All the electrons of non-hybridized and hybridized are open to this
- The electrons in the π bonds are not restricted to a certain C atom. They can move ^{to} every C atom through the chain. So this has de-localized π electron system.
- All the electrons of non-hybridized and hybridized orbitals are open to this π bond cloud.

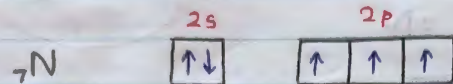


These two orbitals cannot overlap as they are not in sufficient distance in between them. So there isn't a de-localized π electron system.

- So we say the the carbon chain (A) has Conjugated Double bonds while the carbon chain (B) has Isolated Double bonds. (Because some of the π orbitals in chain (B) can't interact with each other.
- So the molecules which belong these two carbon chains are quite different.

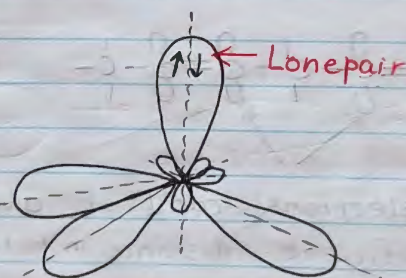
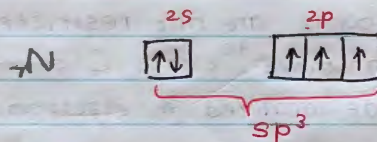


Now let's find the hybridization of the (N) atom in aniline molecule.

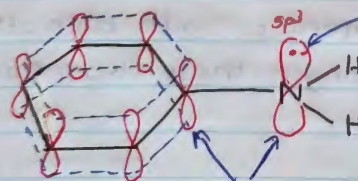


Does this N atom need any excitation for the hybridization of orbitals?

N needs three half filled orbitals for the hybridization. This ground stage atom satisfy this condition. Therefore no need of excitation. The ground stage electron configuration is enough for the hybridization.

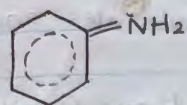


In the arrangement the C-N bond can rotate. So it can change its orientation.



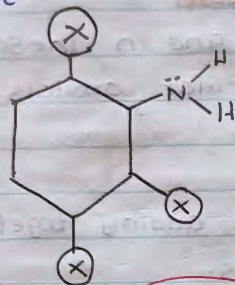
These electrons can move around the C ring and come back to the first place.

These two can also overlapped.



delocalized electron system

When the C-N bond rotates it get conflict with the



When the C-N bond rotates it get conflict or get interrupted by the extra groups ~~with~~ (bulky group) (in ortho, meta, para points).

We call it steric hindrance.

Steric Hindrance.

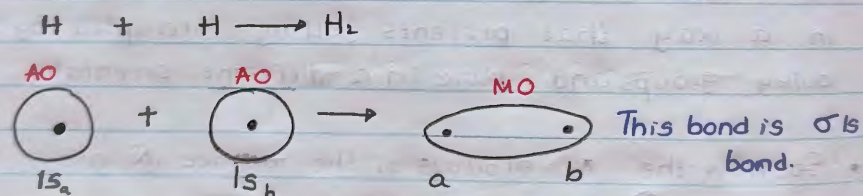
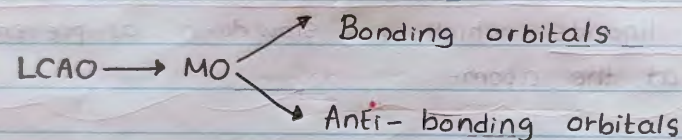
Steric Hindrance at a given atom in a molecule is the congestion caused by the physical presence of the surrounding ligands, which may slowdown or prevent reactions at the atom.

- To avoid this Steric hindrance C-N bond rotates in a way that prevents getting interrupted by the bulky groups and place in a different orientation.
- So now the π orbitals of the ~~adja~~ N atom and the adjacent C can't get overlapped.

- To understand the covalent bonds in molecules and atoms.
 - ① LCAO (Linear combination of Atomic Orbitals)
 - ② Molecular orbitals (MO)
- are very important.

MOLECULAR ORBITAL THEORY

- In atoms occupy atomic orbitals and in the same way electrons in a molecule occupy molecular orbitals, just as electrons in an atom.
- The molecular orbitals arise from adding together atomic orbitals or the wave functions.
- The LCAO creates two types of MOs.

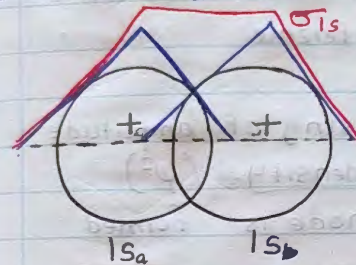


- This is the two way of two orbitals can overlap with each other linearly.
- Also the σ bond is cylindrically symmetrical around the linear bond axis. So the electron density is higher among the two atoms around the bond axis.
- Also there is no nodal along the bond axis.
(If In a nodal plane the electron density is zero. As there's no ~~node~~ such a plane, there's no region ~~at~~ which the electron density / the probability of finding an electron is zero).

- In here there are two possible ways of mixing the wave functions.

- ① Constructive interference.
- ② Destructive interference.

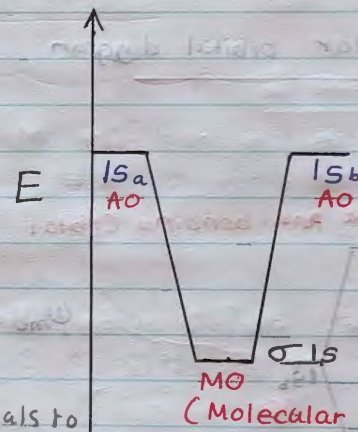
- Let's say, in Constructive Interference.



The two wave functions are in phase they can interact with each other. That means they can interfere constructively, the amplitude increases where they overlap. This gives enhance

probability of electron density between the nuclei.

So the energy of a bonding orbital is decreased compared to the atomic orbital



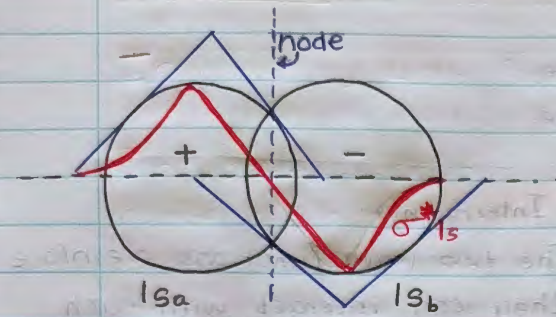
This is equal to
↓
constructive interference

$$\psi_{mo} = \psi_{1s_a} + \psi_{1s_b}$$

← Here the energy amount is very lower than in the atomic orbitals.

So it is very stable compared to the free atoms.

- In Destructive Interference,

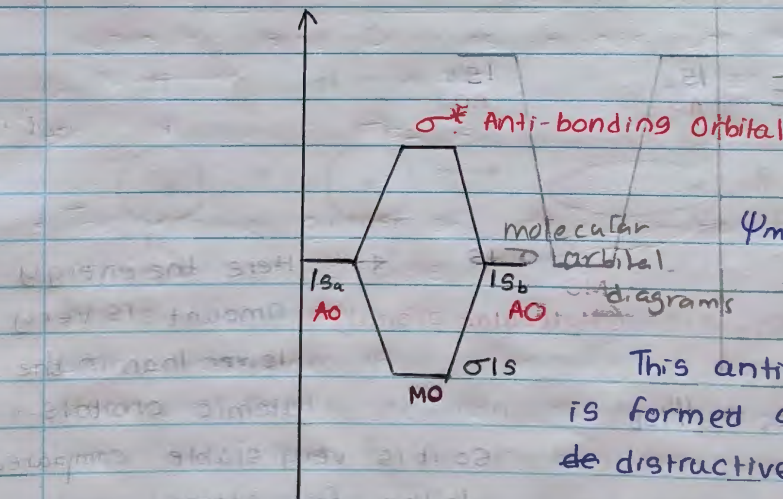


The amplitude is decreases where they overlap, because of their destructive interference

What is important here is, decreasing of amplitude translates to a diminished probability density. (ψ^2) between the nuclei, and a node is formed between the two nuclei.

This bonding is known as $\sigma^* 1s$ or anti-bonding orbital which is higher in energy.

molecular orbital diagram

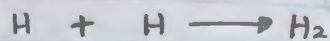


$$\psi_{mo} = \psi_{1s_a} - \psi_{1s_b}$$

This anti-bonding orbital is formed as a result of destructive interference.

- When two atomic orbitals overlap always you get Bonding Molecular Orbital, $\sigma^* 1s$ (Antibonding orbital)

Now let's try to understand the formation of Hydrogen molecule. In terms of molecular orbitals.



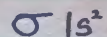
Creating a ' H_2 ' molecule formed by the combination of H atoms. The bonding in H_2 molecule using molecular orbitals.

- How do we arrange the electrons in atom or an orbital?
First we have the electrons to the lowest energy level then if the lowest energy level is filled, then if the next energy level is empty, then the electrons move to the next energy level.
- We always have to distribute electrons from the lowest energy level to the highest energy level.

Special features

- * The electrons in the bonding orbitals stabilize the system. But the electrons in the anti-bonding orbitals destabilize the system. (Doesn't form a bond)
(That's why H exists as molecules in nature instead of as atoms)

So we write the electron configuration as,



- * Predict the stability of the molecule (is it stable or not) it is related to bond order.

Bond Order

Introduced by Linus Pauling.

Bond order is defined as the difference between the number of bonds and anti-bonds. The bond number itself is the number of electron pairs between a pair of atoms.

Bond order always relates to the stability.

$$\text{Bond order} = \frac{1}{2} \left[\text{Bonding electrons in } \text{---} \text{ Number of bonding orbital} \text{---} \text{ @ } \text{---} \text{ in the anti-bonding orbital} \right]$$

$$H = \frac{1}{2}(2-0)$$

$$= \frac{1}{2} \times 2 = 1$$

When the bond order is higher, the bond is stronger.

He₂

Day 11
22/07/22

- The atomic number of He is 2. So the electron configuration of He is 1s².
- Let's assume that two (He) atoms form a He₂ molecule, there are four electrons shared between the two atoms.
- The electrons are always ~~to be filled to the~~ filled to the lowest level. But in here in He atoms lowest level is already filled.

- Now we can see that when the He_2 molecule is formed the two electrons in the bonding orbitals are stabilize the molecule at the same time two electrons in the anti-bonding orbital destabilize the system. Therefore these two lowest stabilities are compared to the atom.

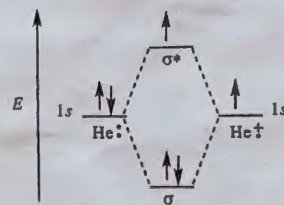
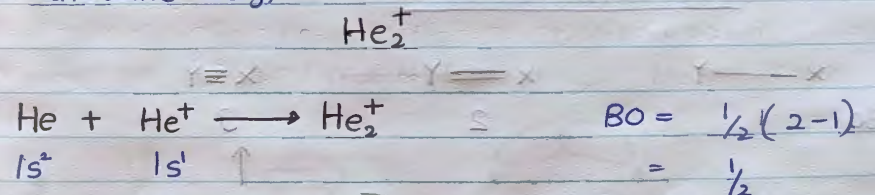


- The energy of the anti-bonding orbital is very high. That's why it destabilize the system. So molecule of He_2 never exists.

$$\text{BO} = \frac{1}{2}(2-2) = 0$$

- Bond order is '0' means that there is no bond formation. So, therefore ~~mono~~ He exist in nature as a mono-atomic form.

- In the same way,



In the He_2^+ molecule-ion, there are only three electrons to be distributed, two from a neutral He atom and one from a positively charged He ion with only one electron and two protons. Two of the electrons go into the low energy bonding molecular orbital, but only one goes into the high energy antibonding molecular orbital. The net result is a lowering of energy of the system, resulting in a bond. The He_2^+ molecule-ion does exist.

Bo ↑

Some criteria that we can discuss about a bond between two atoms.

Let's take $H-H$ as an example.



- Bond distance
- Bond dissociation energy
- Bond strength.

All three of the above are about bond order.

* As the Bond Order ↑

• bond strength → ↑

⇒ This means according to the bond order bond strength also rise up.

When the strength of a bond is high we need more energy to break the bond.

• bond distance → ↓



Bo

1

2

3

Here bond order is 3. This means the bond has more electrons. It is 6 electron between the two nuclei. Therefore negative charged bond cloud can attract the two positively charged nuclei to the centre. As a result the bond becomes shorter.

So we can say,

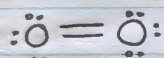
$$\text{Bond distance} \propto \frac{1}{Bo}$$

- bond dissociation energy ↑

(The BDE is the Energy required (an endothermic process) to break a bond and form two atomic or molecular fragments, each with one electron of the original shared pair.)

ex-: Comparing the ~~dis~~ bond dissociation Energy of O_2 and N_2

$$E_{D_{O_2}} < E_{D_{N_2}}$$

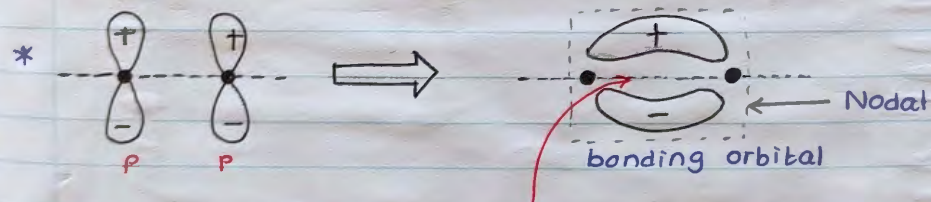


$$B_o = 2$$

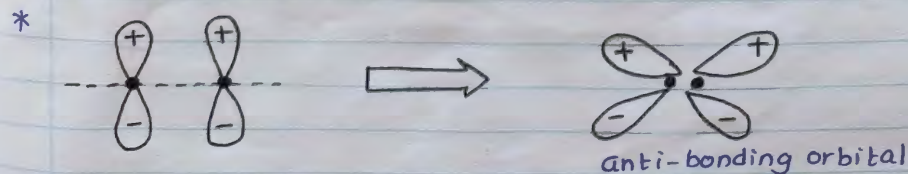


$$B_o = 3$$

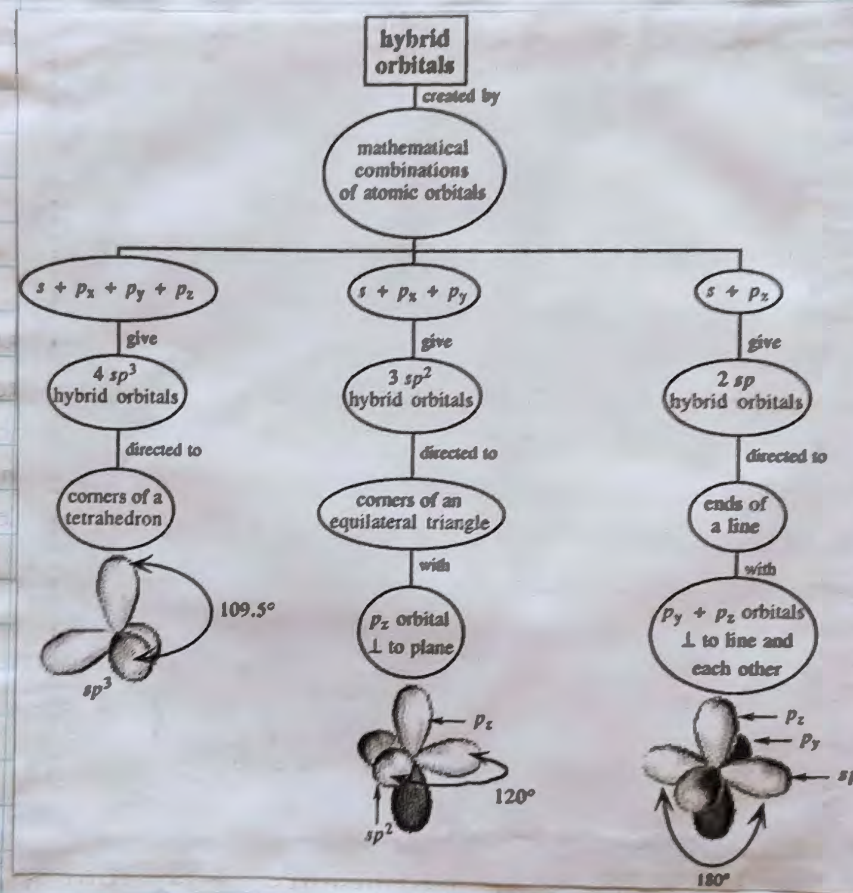
The number of electrons that shared between two N atoms is greater than the number of electrons that shared between two O atoms. So the attraction between 2 N atoms is higher than the attraction between 2 O atoms. So there is a higher amount of energy is need to break $N \equiv N$ bond much than $O=O$ bond.

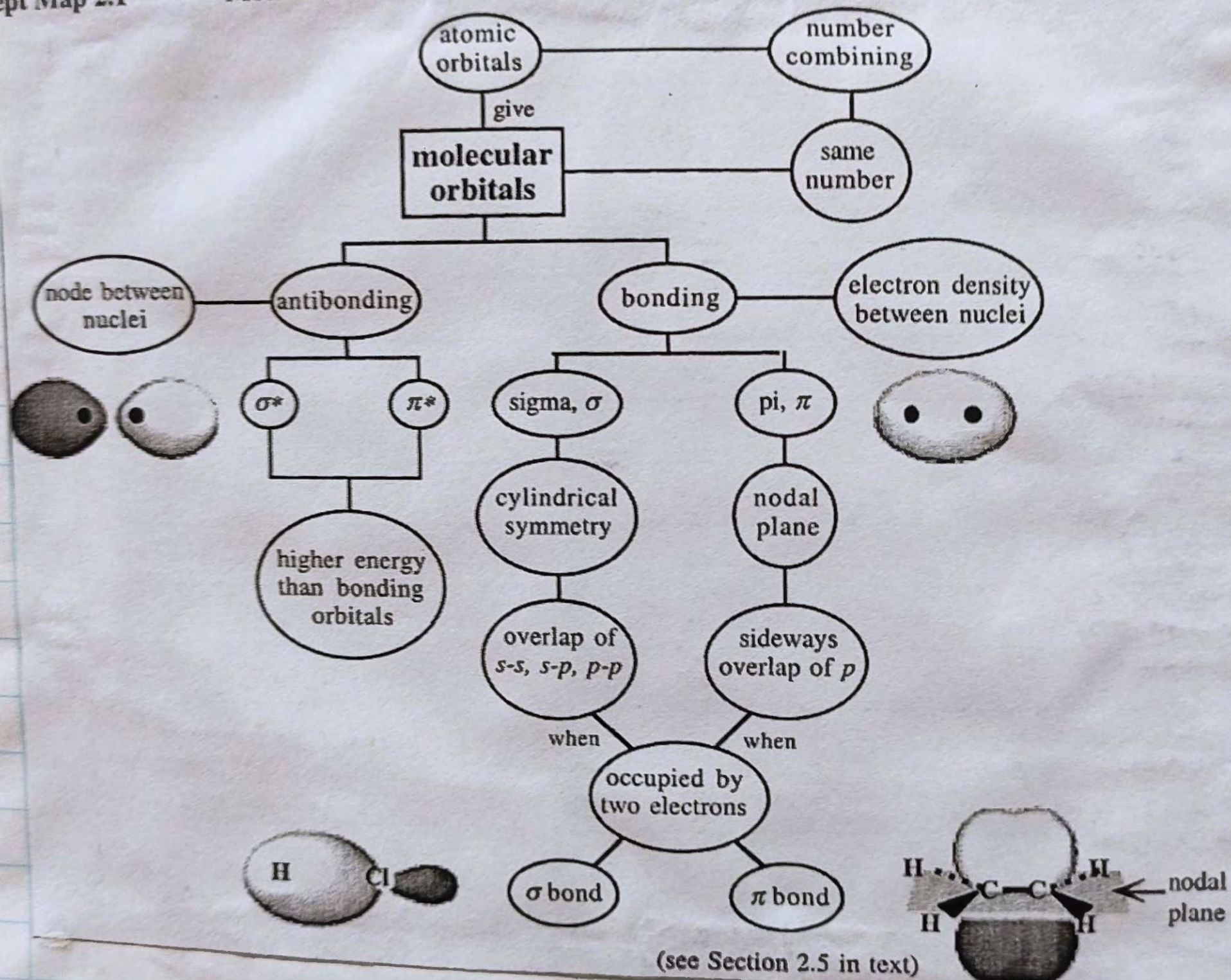


We can't find the π electron cloud/density in this region (along the longitudinal intermediate axis) But ^{there} it is still similar probability of finding the electron above and the below of the intermediate axis.



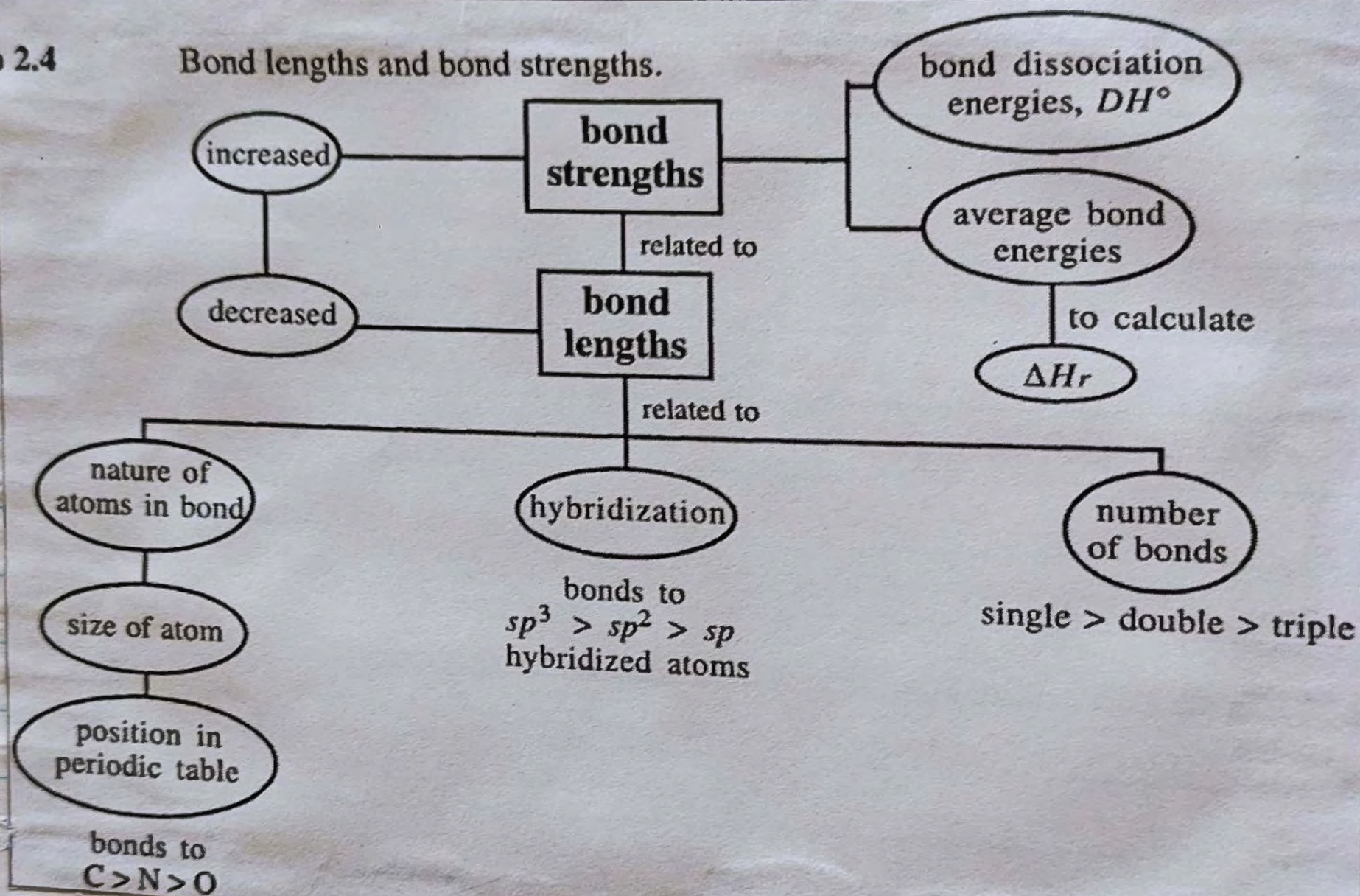
There is bond forming.





2.4

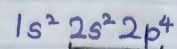
Bond lengths and bond strengths.



hetero $O=C \rightarrow$ The energy levels of two molecules orbitals are complicated. So the initial energy levels are not in the same of atomic molecules. (O₂, N₂, C₂, Li₂ etc.)

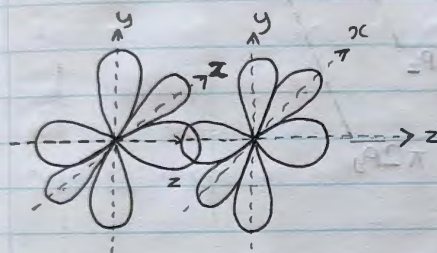
* Let's consider about the O.

Ground stage



There are 6 electrons in the valance electrone shell.

Not like in the previous incidents we took, here there ^{are} ~~is~~ p orbitals ~~are~~ also here.



• Here p_y and p_z orbitals can form only π bonds by overlapping sideways, while the p_z can form ~~a~~ a σ bond by overlapping along the intermediate axis, linearly.

• Let's see how these orbitals arrange in the space in a way of low energy.

• The relative energies of $\sigma 2p_z$ orbitals are compared to the $\pi 2p_x$ and $\pi 2p_y$ orbitals, which depends on the Z values of the atom (Z - atomic number)
(Z - nuclear charge.)

If, $Z < 8$

the condition is

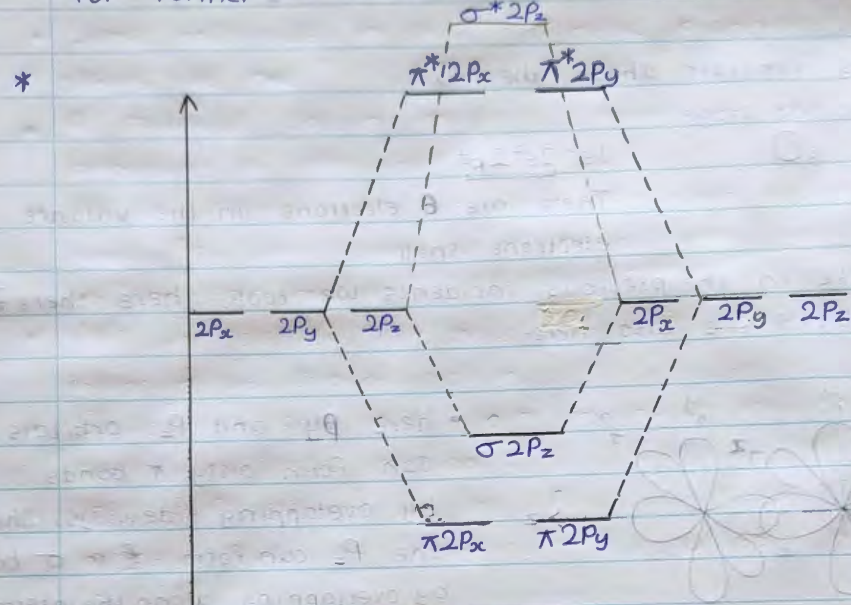
$$\pi 2p_x = \pi 2p_y < \sigma 2p_z$$

If, $Z \geq 8$

$$\sigma 2p_z < \pi 2p_x = \pi 2p_y$$

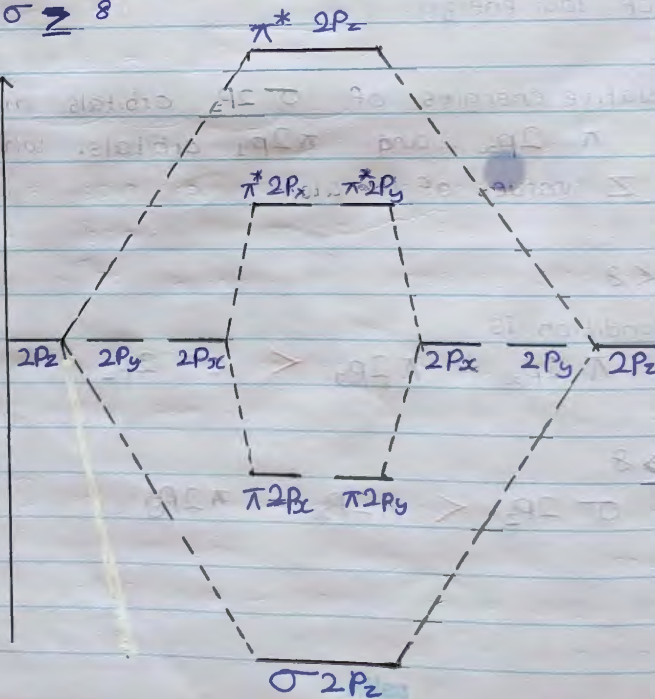
1st Condition.

* Now let's try to draw a molecular orbital diagram for former ~~n~~ Homo-nuclear diatomic molecules.

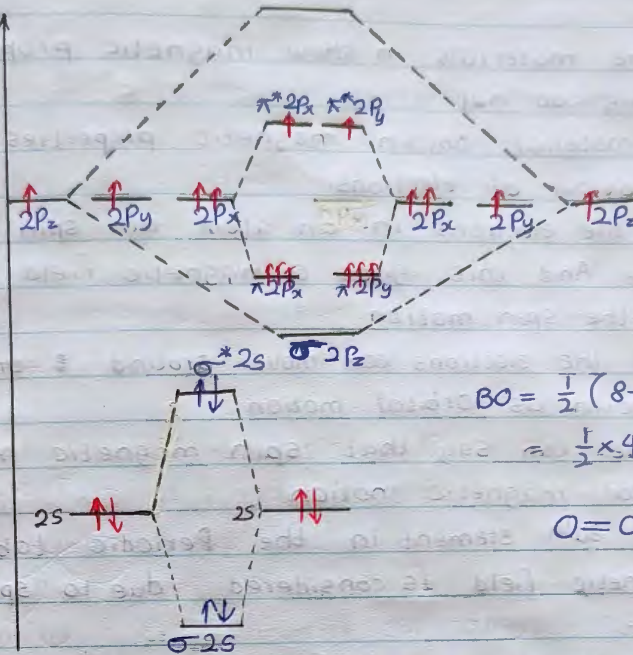
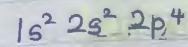


2nd Condition

* When $\sigma \geq 8$



O_2 molecule



$$BO = \frac{1}{2}(8-4)$$

$$= \frac{1}{2} \times 4 = 2$$

$$O=0$$

- Oxygen molecule has two unpaired electrons.
- How do we experimentally prove that Oxygen molecule has two unpaired electrons?

MAGNETIC PROPERTIES OF MATERIALS

- Some materials do show magnetic properties while others do not.
- A material obtains magnetic properties due to the behaviour of electrons.
- All the electrons in an atom can spin around its own axis. And this gives a magnetic field. This is known as the spin motion.
- Also the electrons can move around π orbitals. So this is known as orbital motion.
- So we can say that spin-magnetic motion and orbital-magnetic motion.
- Up to 30th element in the Periodic table, the total magnetic field is considered due to spin motion only.
- Let's see how to calculate the spin-magnetic moment of an electron.

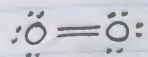
$$\mu_s = \sqrt{n(n+2)}$$

μ_s = spin-magnetic moment measured in Bohr Magneton (BM)

n = The number of unpaired electrons

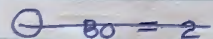
- The behaviour of a matter in the presence of a magnetic field can give a way to classify those matters.
 - ① Diamagnetism $\rightarrow$ no unpaired electrons.
 - ② Paramagnetism $\rightarrow$ weakly repelled from a magnetic field.
 - ③ Ferromagnetism $\rightarrow$ one or more unpaired electrons attracted to a magnetic field.When the electron number of electrons is higher, the attraction increases.

- Now let's see what type of magnetism that shows by O_2 molecules.



As all the valence electrons are paired, there are no unpaired electrons can't be found. So, ~~we can~~ it is obvious that O_2 molecule is diamagnetic.

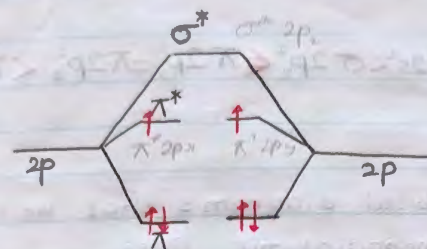
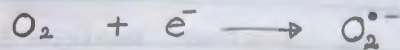
- But according to the molecule orbital theory, O_2 have two unpaired electrons. ~~So~~ But we considering the Lewis structure it is diamagnetic.
- But ~~the~~ it has proved that O_2 molecule is Paramagnetic. This means that this molecule ~~has~~ should have at least one unpaired electron.
- Oxygen can form O^{2-} (normally) oxi-anions.
- If we give an electron to the molecule,



Let's consider about the bond orders now.



Day 12
26/02/2022



Molecular orbital diagram for ground state O_2 molecule.

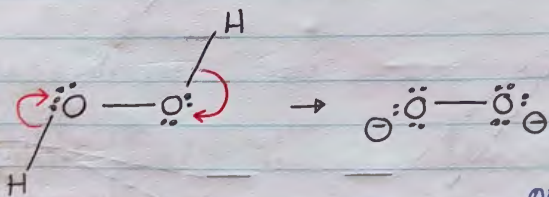
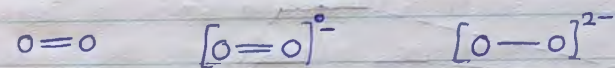
Let's calculate the bond order.



$$BO \quad O_2 > O_2^{\cdot -} > O_2^{2-}$$

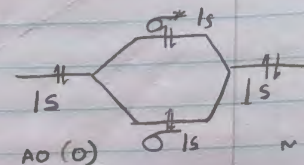
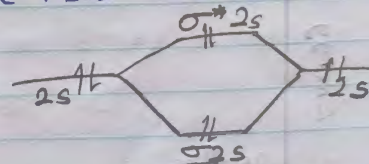
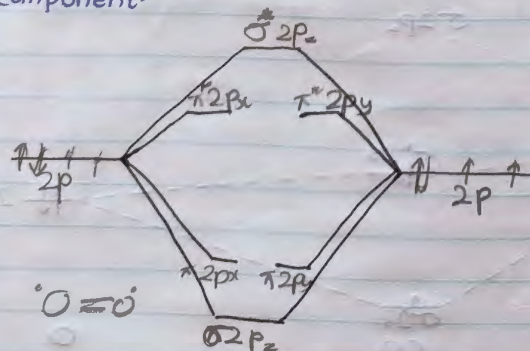
$$\text{Bond Strength} \quad O_2 > O_2^{\cdot -} > O_2^{2-}$$

$$\text{Bond length} \quad O_2 < O_2^{\cdot -} < O_2^{2-}$$



here oxygen have more electrons in π^* anti-bonding orbitals. the bond

becomes weak. That's why the H_2O_2 is a more unstable component.

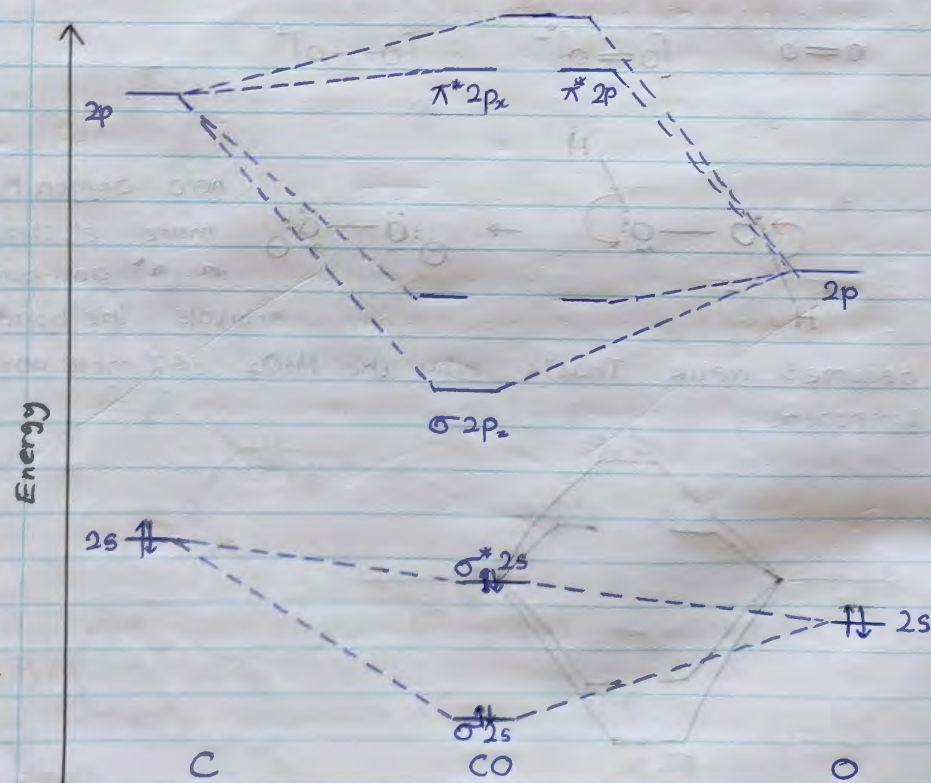
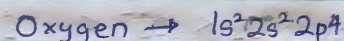
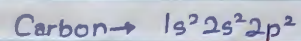


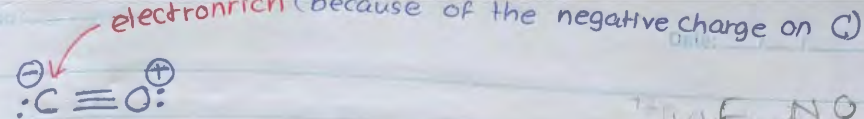
- How can we write the electron configuration of the a molecule by using the molecular-orbital diagram.

$$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < \pi 2p_x^2 = \pi 2p_y^2 < \pi^* 2p_x^2 = \pi^* 2p_y^2$$

- All those molecular orbital diagrams that we considered above were about monodiatomic ones. But, now let's see how to draw a molecular orbital diagram for a heterodiatomic molecule. (CO, HF, NO ---- etc.)

- Let's consider the CO (Carbon Monoxide) molecule. Electron configurations

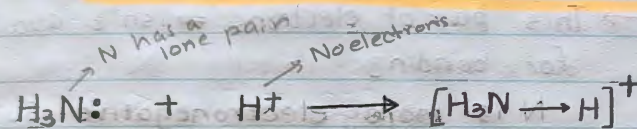




This is a diamagnetic molecule. So the bond is much stronger. There are two unpaired electrons which don't used for bonding.

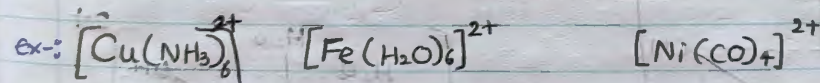
$$B.O = \frac{1}{2}(6-0) = 3$$

CO-ORDINATE COMPOUNDS



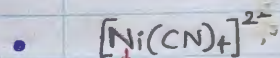
* The bond is formed by one atom sharing the electron pair. The other atom has a completely empty orbital. These type of bond is called as co-ordinate bond or dative bond.

* Now let's consider about co-ordinative complexes.

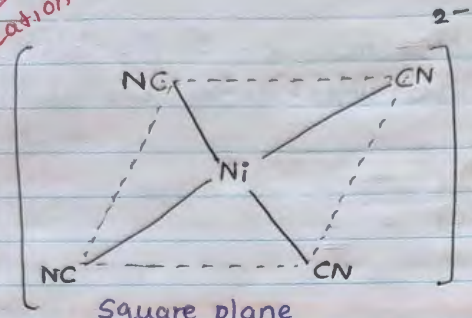


When the co-ordinate bond forms. That entity that gives the pair of electrons is called the ligand.

ex-: In $[\text{Cu}(\text{NH}_3)_6]^{2+}$ the ligand is $:\text{NH}_3$
 In $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ the ligand is H_2O
 In $[\text{Ni}(\text{CO})_4]^{2+}$ the ligand is CO



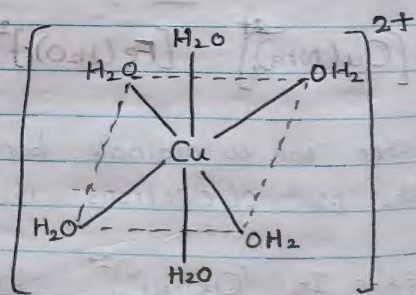
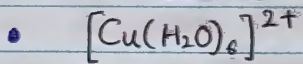
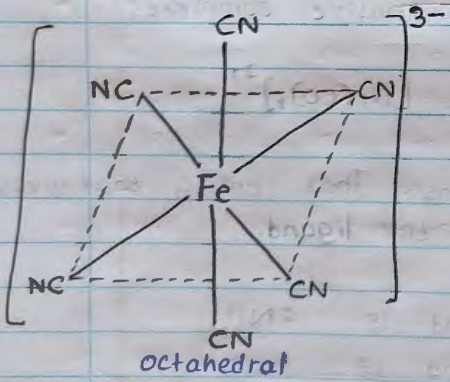
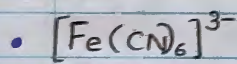
dsp² hybridization

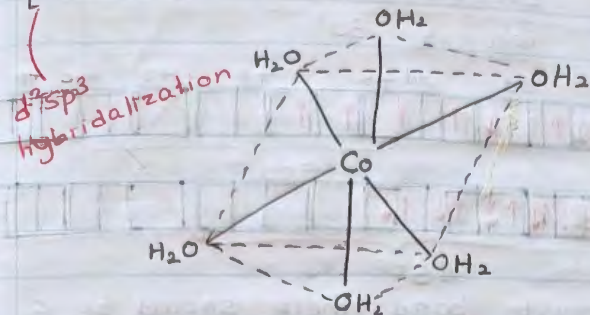
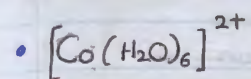


* Once the co-ordinate bond forms it is similar to a covalent bond.

lone pair of electrons $\leftarrow \text{C} \equiv \text{N} \rightarrow$ This pair of electrons isn't available for bonding.

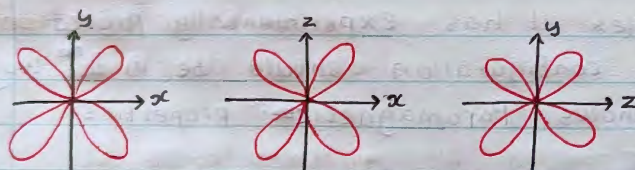
This is available for bonding. $\text{H}-\text{N}-\text{H}$ is highly electronegative. So, the pair of electrons is tightly bound to the N atom. That's the reason for the N atom isn't available for making bonds.





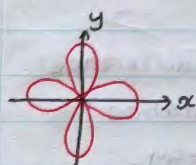
- If you consider the electron configuration of transition metals they use 'd' orbitals.
- Considering about the d orbitals.

d_{xy} d_{xz} d_{yz}



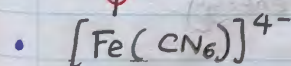
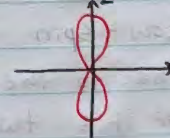
d_{xy} , d_{xz} and d_{yz} orbitals distribute among the axis.

$d_{x^2-y^2}$



these orbitals distribute just along the axis.

d_{z^2}



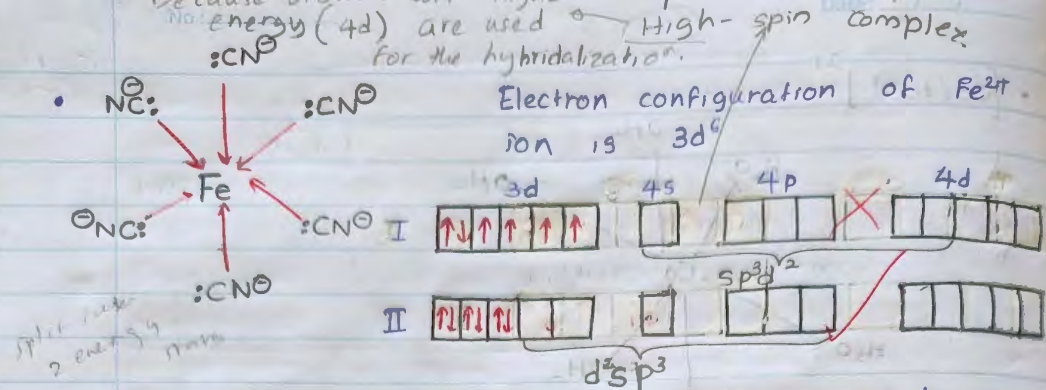
valence shell of electron configuration of Iron is

$3d^6 4s^2$

$$\begin{aligned} x + (-6) &= 4 \\ x &= -4 + 6 \\ x &= +2 \end{aligned}$$

So, to form the Fe^{2+} ion the ground stage Fe atom should remove $4s^2$ electrons.

So, the electron configuration of Fe^{2+} is $3d^6$.

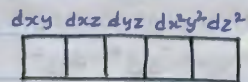


- To make a co-ordinate bond there should be 6 empty orbitals.
- To form Fe^{2+} ion there can be two possibilities of the configuration of the Fe^{2+} .
- So it is obvious that IInd possibility should be the electron configuration for the Fe^{2+} to make co-ordinate bond with 6 CN^- ions.

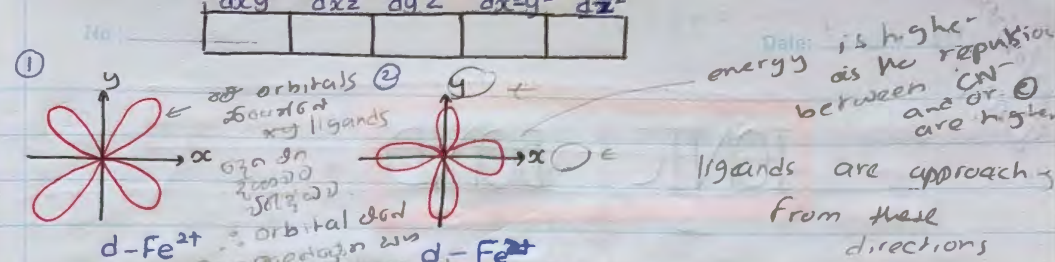
Because, by measuring the magnetic properties of $[\text{Fe}(\text{CN})_6]^{4-}$ complex it has experimentally proved that that the configuration should be in IInd way.

- This complex shows magnetic properties (All electrons are paired. No unpaired electrons).
- To justify that that statement, it is clear that the correct electron configuration of Fe^{2+} should be the hybridization ' d^2sp^3 '
- We call this as a Spin-paired complex or Low-spin complex
- What is the simple reason for taking the difference of this two ways of electron configuration?

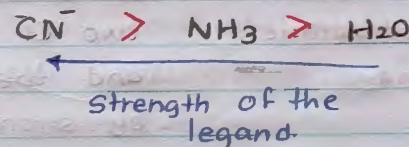
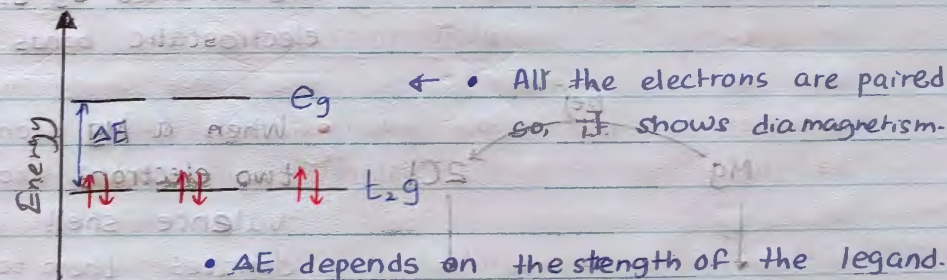
All d orbitals are not in same energy.



- If all the orbitals are in same energy it is known as degenerated orbitals.
- So the simple answer for the above question is the d orbitals are not degenerated.



- The ligands should come with the pair of electrons give them to the empty orbitals
- In the second diagram ligands are approaching from the directions of the same way of orbitals. So the repulsion between CN^- ions and electrons in orbitals are higher.
- In the first diagram orbitals are directed in opposite way from the direction of the approaching of CN^- ions. So, the repulsion between the electrons in the orbitals and CN^- ions are minimum.



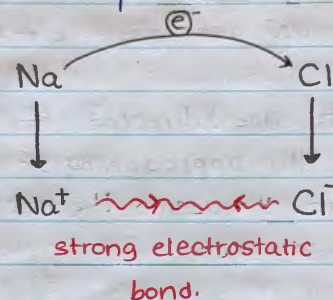
- * dx^2-y^2 and dz^2 point along the axis
- * dxy , dyz , dxz point at in between the axis.

27/02/2021
Day 13

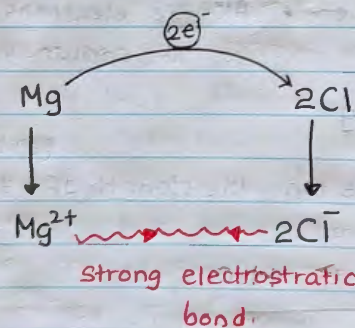
No. 1s²2s²2p⁶3s¹

IONIC BOND

- ★ Ionic bond is a type of bond that involves the electrostatic attraction between oppositely charged ions, or between two atoms with sharply different electronegativities, and is the primary interaction occurring in ionic compounds.



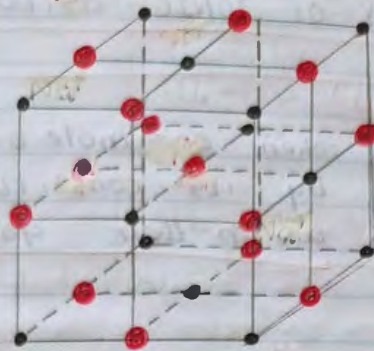
- When a Na atom gives an electron to a Cl atom, Na⁺ and Cl⁻ ions are formed. And then those positively charged and negatively charged ions are attracted by strong electrostatic bonds.



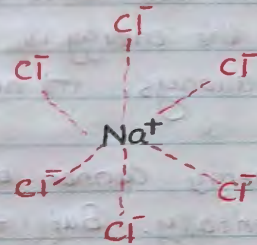
- When a Mg atom removes two electrons from its valence shell and donates them to two Cl atoms, Mg²⁺ and two Cl⁻ ions are produced and they get attracted by strong electrostatic bonds.

- These ions are arranged in a geometrical order to form an Ionic Lattice.
- In all lattices each positively charged or negatively charged ion is surrounded by a maximum number of oppositely charged ions, forming crystals.

ex -



● → Na^+
● → Cl^-



Each Na^+ atom is surrounded by 6 Chloride ions.

Properties of ionic compounds

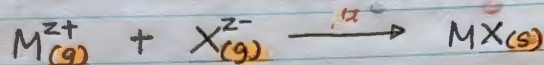
- ① They form crystals.
- ② They have higher melting points and higher boiling points.
- ③ They have higher enthalpies of fusion and vaporization than molecular compounds.
- ④ They're hard and brittle.
- ⑤ They conduct electricity when they are dissolved in water.
- ⑥ They are good insulators.
- ⑦ Have a higher polarity.

STABILITY OF IONIC COMPOUNDS

Lattice Energy (U)

Energy liberated when one mole of an ionic crystal is formed by its constituent cations and anions which are in their gaseous state.

ex:-



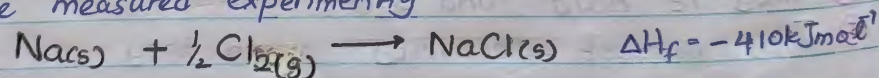
- As the energy is given out while forming an ionic compound by its ions, the energy is always released so the value ' U ' is always $(-)$ negative.
- This amount of Lattice energy cannot be directly measured by experimentally. But this can be measured by indirectly by **Born Haber Cycle**.

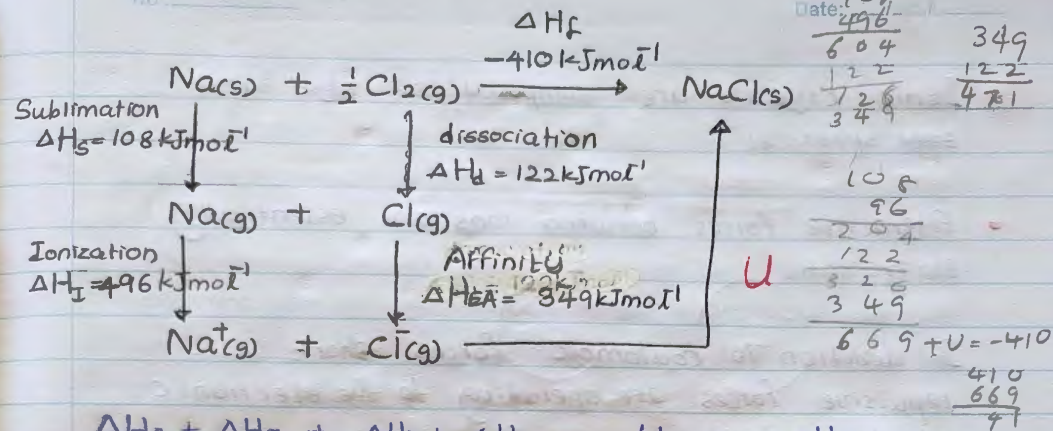
Now let's see how to use the Born Haber method to determine the lattice energy.

ex:- $NaCl_{(s)}$

Enthalpy of formation (ΔH_f)

- * The enthalpy of formation of $NaCl_{(s)}$ crystalline can be measured experimentally





$$\Delta H_s + \Delta H_i + \Delta H_d + \Delta H_{EA} + U = \Delta H_f$$

$$108 \text{ kJ mol}^{-1} + 496 \text{ kJ mol}^{-1} + 122 \text{ kJ mol}^{-1} + 349 \text{ kJ mol}^{-1} + U = -410 \text{ kJ mol}^{-1}$$

$$U = -763 \text{ kJ mol}^{-1}$$

- Melting point of sodium chloride depends on the lattice energy.

Lattice Energy $\propto$ Melting point

Lattice Energy $\propto$ solubility

- The lattice energy of crystalline ionic compound can be theoretically calculated by the equation.

$$U = \frac{N_A M Z^+ Z^-}{4 \pi \epsilon_0 r} \left(1 - \frac{1}{n} \right)$$

Born - Lande Equation

$N_A \rightarrow$ Avagadro constant $6.02 \times 10^{23} \text{ mol}^{-1}$

$M \rightarrow$ Madelung constant

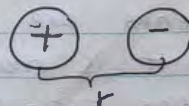
(This relates to the geometry of the crystal)

$Z \rightarrow$ Charge number

$\epsilon_0 \rightarrow$ Permittivity of free space $8.854 \times 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1}$

$r \rightarrow$ Inter ionic distance

$n \rightarrow$ Born exponent



- Ionic crystals are composed of rigid and spherical.
- Attractive forces between ions are essentially electrostatic.

In addition to coulombic forces, short repulsive forces are operating to the electronic clouds.

$$U \propto \frac{1}{r+r'} \quad U \propto Z^+ Z^-$$

Compound	Theoretical U kJmol ⁻¹	Calculated U kJmol ⁻¹
NaCl	766	788
highly ionic NaI	686	699
AgCl	768	890
Act as a covalent character AgI	736	867

Doesn't have complete ionic character

which cause for the higher

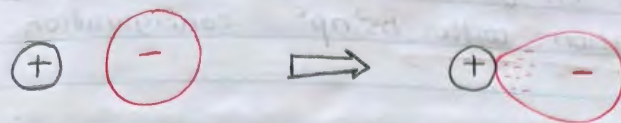
By these calculations we can see that there's no much difference between the theoretical value and the calculated value of U of NaCl and NaI.

Evidence: Not soluble in water

- But there's a considerable difference can be seen in AgCl and AgI of those two values.

Let's see what is the reason for this. In Born-Haber method we assume that the whole amount of elements are completely ionized. But in theoretical method $\pm$ charges are considered only. So the above assumption can't be used to AgI and AgCl.

FAJANS RULES



When the anion is much larger compared to the cation the electron cloud is much more diffused from the nucleus.

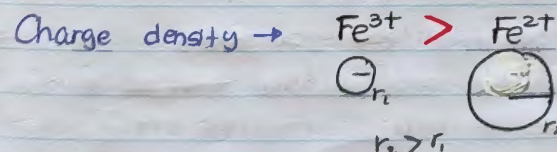
~~The cation~~ The electron concentration around the cation is much more attracted to the nucleus, generally

$$r^- > r^+$$

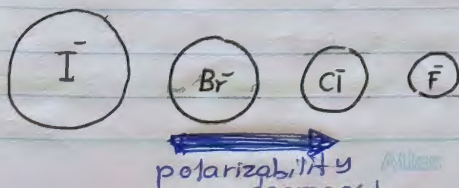
The polarizing power of the cation can attract the electron cloud because of the positive charge. Polarizing power is the extent to which a cation can polarize an anion / distortion of a spherical ~~ion~~ symmetric electronic cloud to an unsymmetric cloud.

Polarizability $\rightarrow$ extent to which ~~an~~ an anion can be polarized

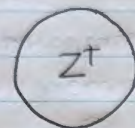
- ① Smaller cations and higher charged number and a higher polarizing power $\rightarrow$ (Then charged density becomes higher)



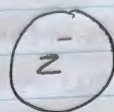
- ② Larger anions can be easily polarized.



③ For the cation with same size and charge, the one with $(n-1)d^{10}ns^0$ has higher polarizing power than the cation with ns^2np^0 configuration.



$(n-1)d^{10}$



ns^2np^0

Adm
Cations
have
some
polarization

I^- much bigger
 F^-
 Na^+ more
covalent

more ionic
more

Q1) NaI, NaF Which is more water soluble and have higher melting point?

more covalent
MgCO₃, CO₃²⁻ Which has lower thermal decomposition temperature.

highly polar
Mg²⁺



Day 14

05th of Feb

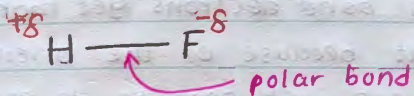
DIPOLE MOMENT

ද්විධ්ರುವ අවස්ථාව

* Let's take HF molecule as an example



- Fluorine is more electro-negative than Hydrogen
- So the bond pair of electrons ~~are more~~ behave more closer to the F. So, the F ~~is~~ electron density is slightly increases in F. Because of that F gains δ^- charge.



- And according to that H atom gains δ^+ similar to the F atom.

- So the HF molecule has a positively charged ^{in one} end as and also a negatively charged end. Therefore, the we call the bond between H and F as a polar bond.

* Another way to represent the polarity of a molecule is drawing an arrow ~~that~~ the arrow head directed to the more electro-negative atom.

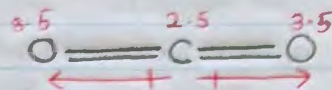


* If we consider the bond between C and H atoms, it is not a polar bond. Because there's no electro-negativity difference between C and H.

Polar Bond

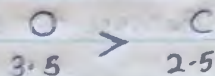
A polar covalent bond is a bond formed when a shared pair of electrons are not shared equally.

* CO_2 as a polar molecule?



Here there are two O atoms have bonded with the C atom.

Electronegativity



So, what happens is bond electrons get more attracted to the O atoms. But because of the orientation of the O atoms are just opposite to each other, the polarity same polarity to the two opposite directions get cancelled by each other.

Therefore CO_2 is not a polar molecule.

* To say a molecule to be polar, the positive and negative centers of the molecule must not coincide.

* The degree of the polarity of a molecule is measured by its dipole moment (μ).

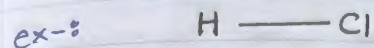
$$\mu = Qr$$

unit Debye

$r \rightarrow$ distance between the two poles

$Q \rightarrow$ charge at the end of the two poles dipole, (either one)

$$1\text{D} = 3.43 \times 10^{-30} \text{ C m}$$



$\text{Na}^+ - \text{Cl}^-$
this is not partially polarize
It is highly polarized

$$D = 1.03 \text{ D}$$

$$r = 1.36 \text{ \AA}$$

$$Q = ?$$

$$1 \text{ \AA} = 1 \times 10^{-10} \text{ m}$$

$$\mu = Qr$$

$$1.03 \times 3.43 \times 10^{-30} \text{ C m} = Q \times 1.36 \times 10^{-10} \text{ m}$$

$$Q = \frac{1.03 \times 3.43 \times 10^{-20} \text{ m}}{1.36}$$

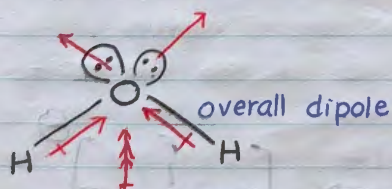
$$Q = 2.52 \times 10^{-20} \text{ C}$$

Charge of an electron is $1.602 \times 10^{-19} \text{ C}$

But according to that the charge of the H atom or the Cl atom of HCl molecule is less than the charge of the electron. This is true because HCl is not a much ionic compound like NaCl. For NaCl you can gain the value of the charge of the electron.

$$\text{Percentage of charge} = \frac{2.52 \times 10^{-20} \text{ C}}{1.602 \times 10^{-19} \text{ C}} \times 100\%$$

$$= 15.8\%$$



LEWIS STRUCTURE

* Let's see how to draw a Lewis structure of a given molecule.

- ① Add the number of valance electrons provided by each atoms.
- ② Add one for each negative charge and subtract one for each positive charge.
- ③ Determine the number of electron pairs by deviding by 2.

ex-: CCl_4

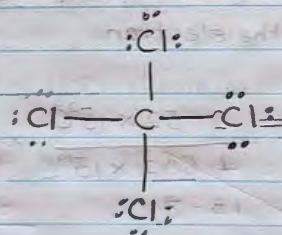
$$\text{① } \text{C} \rightarrow 4$$

$$4 \times \text{Cl} \rightarrow 7 \times 4 = 28$$

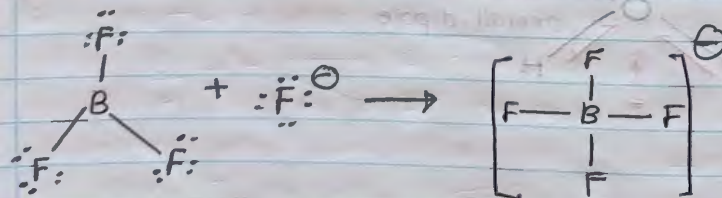
$$\text{Total } 32$$

② There are no charges

$$\text{③ } \frac{32}{2} \rightarrow 16 \begin{cases} \rightarrow 4 \text{ bonds} \\ \rightarrow 12 \text{ lone pairs} \end{cases}$$



I BF_4^-



* In a Lewis structure the number of electrons around each electron should be in the octate state.

In this example F atoms fulfil this requirement but not the B atom. Because it has only 6 electrons around it. Therefore BF_3 is an electron deficient molecule.

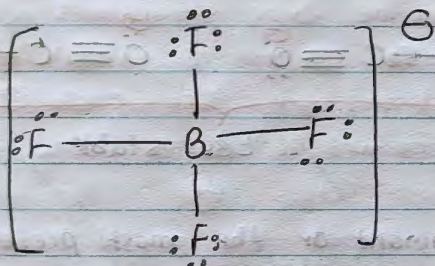
So, the BF_3 molecule can react with $:\ddot{\text{F}}:^- / \text{F}^-$ ion, because it has an extra electron.

$$\textcircled{1} \text{ B} \rightarrow 3$$

$$4 \times \text{F} \rightarrow 7 \times 4$$

$$\textcircled{2} \frac{32}{2} \rightarrow 16$$

$$\textcircled{2} \frac{32}{2} \rightarrow 16 \quad (4 \text{ for bonds } \text{F}^- \quad 12 \text{ for lone pairs})$$



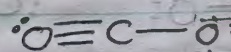
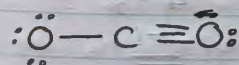
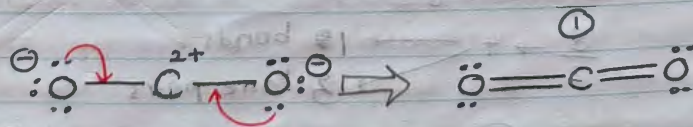
III CO_2

$$\textcircled{1} \text{ C} \rightarrow 4$$

$$2 \times \text{O} \rightarrow 6 \times 2 \rightarrow 12$$

$$\textcircled{2} -$$

$$\textcircled{3} \frac{16}{2} = 8$$



Formal Charge

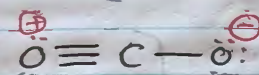
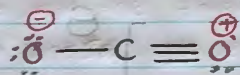
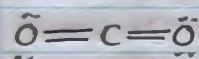
$$\text{Formal Charge} = \left(\text{Number of valence electrons} \right) - \left(\text{Number of bonds to the atom} + \frac{\text{Number of unshared electrons}}{2} \right)$$

ex-: O in CO₂

C in CO₂

$$\begin{aligned} F.C &= 6 - (2 - 4) \\ &= 6 - 6 \\ &= 0 \end{aligned}$$

$$F.C = 4 - (4 + 0)$$



Minimum formal charge.

Less stable.

therefore the dominant or the most preferred structure for CO₂

IV CO

$$C \rightarrow 4$$

$$\frac{10}{2} \rightarrow 5$$

$$O \rightarrow 6$$

$$\frac{10}{10}$$

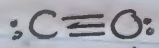
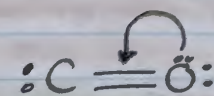
Assuming that O forms a double bond with

$$10 - 2 \rightarrow 8$$

$$\frac{8}{2} \rightarrow 4$$

12 bonds

3 lone pairs



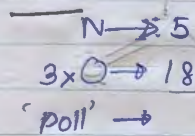
$$4 - (3 + 2)$$

$$4 - 5$$

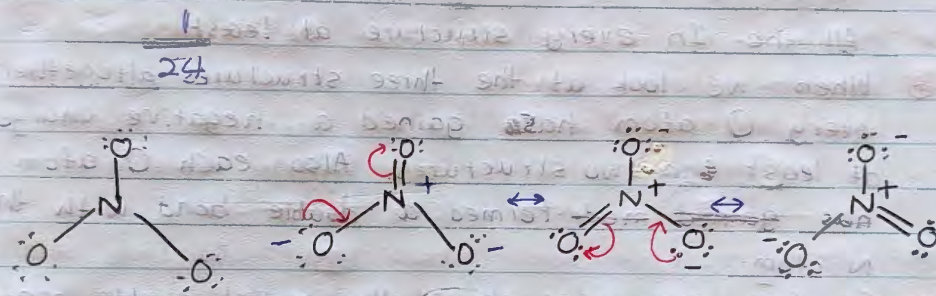
$$6 - (3 + 2)$$

$$6 - 5$$

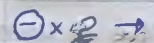
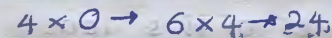
Having a positive charge on a negative (electro negative) oxygen, because it is unstable.



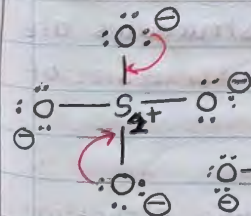
$$\frac{24 - 6}{2} = \frac{18}{2} = 9$$



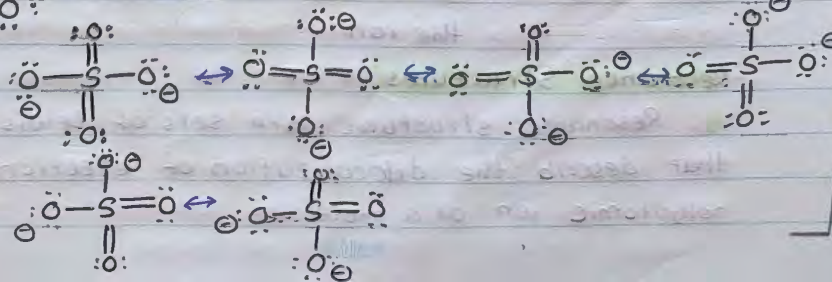
NO_3^- ion has a partial double bond character.



$$\frac{32}{2} = 16$$

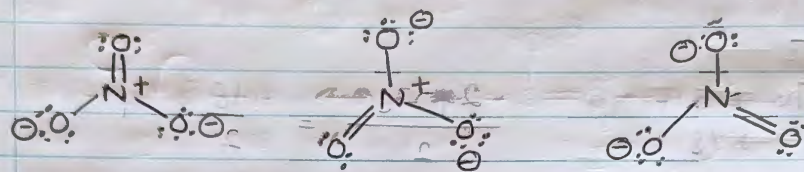


S can form more than 4 bonds. maximum upto six bonds. Therefore,



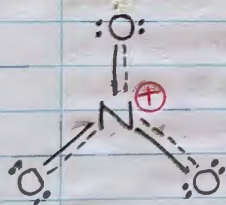
- * Because of the partial double bond character of the NO_3^- ion we use a most appropriate way to represent the Lewis structure of the ion.

- * ~~When~~ ^{Let's} we consider about the three possible structures of the NO_3^- ion first.



- * When we look at the three structures altogether, every O atom has gained a negative charge at least in two structures. Also each O atom has gained made formed a double bond with the N atom.

- * Therefore considering about the partial double bond character of the O atoms, we can represent the Lewis structure as given below.



- * In the three structures of this ion, the electronic structure of the atoms are different in each of them. But positions of the atoms are always constant. Those three structures are known as the **Resonance Structures** of the ion.

Resonance Structures

- * Resonance structures are sets of Lewis structures that describe the delocalization of electrons in a polyatomic ion or a molecule.

* Resonance hybrid of all the three above structures is known as the actual structure of the ion.

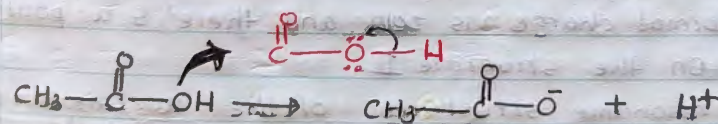
* The resonance structures are very important to determine the stability of an ion or a molecule.

- If an ion or a molecule has many resonance structures that particular ion is more stable.
- If an ion or a molecule has few resonance structures that particular ion or molecule is more less stable.

* When higher the ~~res~~ identical resonance structures, all contribute equally for the stability of the ion or the molecule.

* $\text{CH}_3\text{COOH} \rightarrow$ Acetic Acid / Ethanoic Acid / Ethylic Acid / Vinegar Acid / Methane Carboxylic Acid.

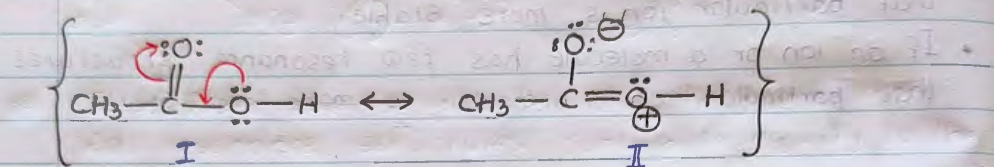
► Acetic Acid is weakly acidic because CH_3COOH dissolves in water forming acetate (CH_3COO^-) and H^+ ions.



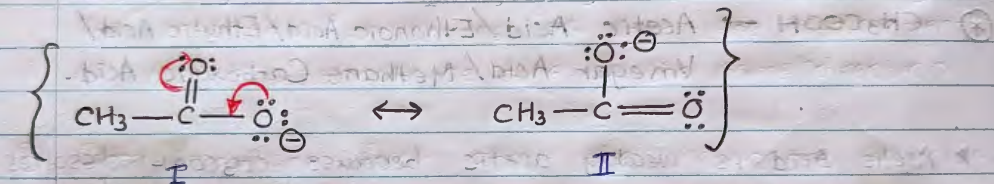
► The reason for the possibility of this reaction is that the product (CH_3COO^-) is more stable than the reactants (CH_3COOH).

▶ Now let's see how we can draw the resonance structures for the acetic acid.

▶ By keeping the number of atoms and the positions of the atoms, constant and changing only the distribution of the valence electrons of the atoms, resonance structures are made.



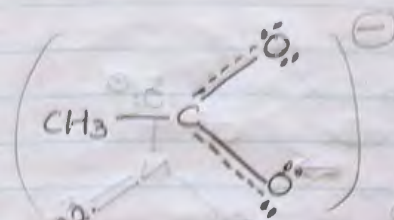
▶ As the same way, we can draw the resonance structures for the acetate ion.



▶ ~~In the~~ When we look at the resonance structures of the CH_3COOH molecule, those two resonance structures are not identical. Because the I structure is neutral as the formal charge is zero and there's a positive charge is on the structure II.

The two resonance structures of the acetate ion are identical. Therefore greater contribution ^{took} of the resonance structure stabilization of the hybrid resonance. Also there is no charge separation in these resonance structure. Also there's a minimum formal charge ~~comparing~~ compared to ~~the~~ the resonance structures of the CH_3COOH molecule.

- So we can represent the resonance hybrid structure of the acetate (CH_3COO^-) as below.



Day 15
12/02/2022

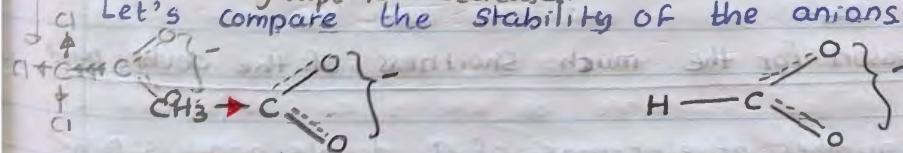


- ⊛ When you dissolve 1 mol of strong acid in water we can gain 1 mol of H^+ and 1 mol of Cl^- are formed. Because they completely dissociate into H^+ and the anion.

- ⊛ But in weak acids this happens the dissociation happens in 1-2% possibility.

- ⊙ What is more acidic between Acetic acid (CH_3COOH) and Formic acid (HCOOH).

Cl has electron withdrawal effect. Therefore the electron density of the carboxylic groups is decreases. Let's compare the stability of the anions

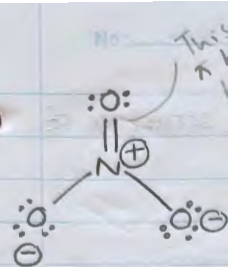


The alkyl group has an inductive effect. This means they can release/push electrons. So,

The H atom has no ability to make inductive effect. So this is much more stable than the CH_3COO^- .

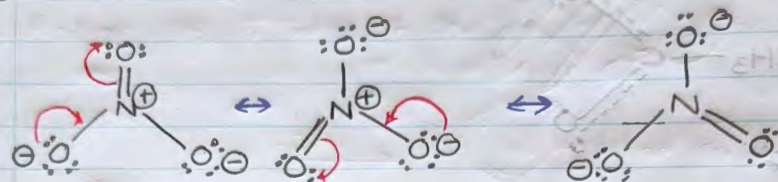
Therefore, we can conclude that (HCOOH) Formic acid is more acidic than (CH_3COOH) Acetic acid.

This withdrawal effect of Cl can make stable. the anion more acidic. So it becomes more acidic than CH_3COO^-



This π bond can be easily broken because it is formed by the overlap of p orbitals.

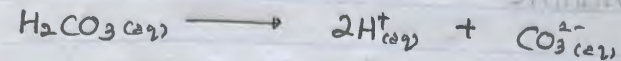
Now let's assign the formal charge of each atom.



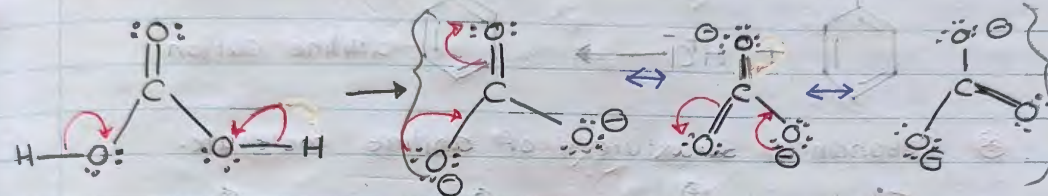
- * These three are stable because they are identical. All resonance structures contribute equally to the stability of the hybrid.
- * Something important regarding formal representing the resonance structures is that the displacement of electrons should be done to the same direction.
- * Other important fact about the resonance structures is, that all the atoms in the resonance should be in the same plane, of the ion or the molecule.
- * When If we consider about the bond lengths of each bond, single bond is ~~much~~ longer the double bond. The reason for the much shortness of the double bond

The number of electrons of a double bond is higher than the a single bond. Because of this electron density in between two atoms, the attraction of the nucleus of two atoms increases. Due to this the two atoms get more closer. So as a result the atoms get closer and the bond between two of them get much shorter. Also we can conclude that a triple bond is more shorter than a double bond.

• H_2CO_3 (Carbonic Acid)

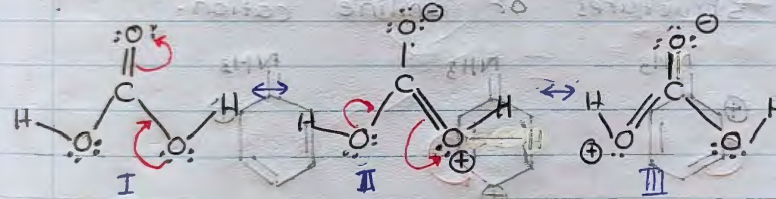


⊛ H_2CO_3 is acidic because CO_3^{2-} is more stable than H_2CO_3 molecule.

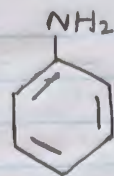


The three resonance structures are more stable.

⊛ Let's try to draw resonance structure for the acidic molecule.

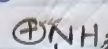
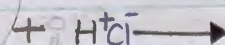
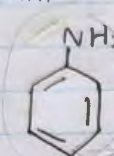


⊛ When we consider about the three resonance structures the first structure doesn't form any formal charge or charge separation. But in other two resonance structures, each one has a positive charge on carbon oxygen. Therefore, only I is stable.



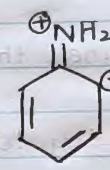
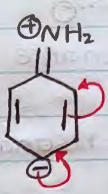
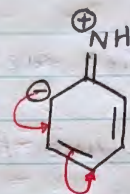
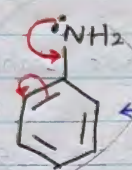
Aniline

* Aniline is a weak base.

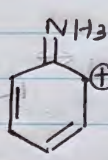
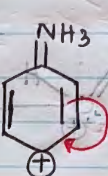
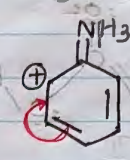
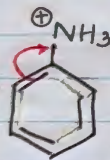


aniline cation

* Resonance structures of aniline molecule.



* Resonance structures of aniline cation.

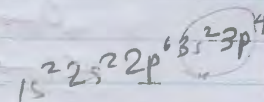
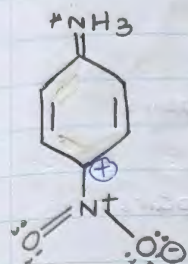


* Aniline molecules resonance structures have charge separation. Also there's a positive charge on electro-negative N atom. Therefore they are not stable.

All structures do not contribute equally to the stability of ion/molecule

* Also, in the resonance structures of aniline cation don't show any charge separations. And, also those are identical. Because of this aniline cation is more stable. Therefore if we dissolve NH_4^+ Aniline in HCl aniline reacts molecules reacts with the H^+ (protons) to form the aniline cations.

* When extra groups are bound to the para point.



By binding NO_2 group to the C cycle the C atom where contacted with NO_2 group gets a positive

charge. Now there are two positive charges (one on N and the other on the N). To keep the positive charges closer higher amount of energy is stored. Because of this this becomes less stable. Therefore this becomes less stable.

As the methyl group is inductive group they push electrons towards the C cycle and stabilize where the p had an electron deficiency the structure become more stable.

• SO_2Cl_2

S $\rightarrow$ 6

2 x O $\rightarrow$ 4

2 x Cl $\rightarrow$ 2

12 $\rightarrow$ 6
2

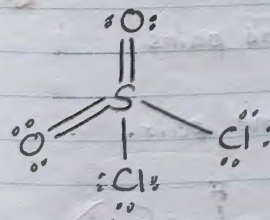
SO_2Cl_2

S $\rightarrow$ 6

O $\rightarrow$ 12

O $\rightarrow$ 14

32
16



tetrahedral.

4 repulsive units

bond pair

lone pair

• SOCl_2

$$\begin{array}{r} \text{S} \rightarrow 6 \\ \text{O} \rightarrow 2 \\ 2\text{Cl} \rightarrow 2 \\ \hline 10 \end{array}$$

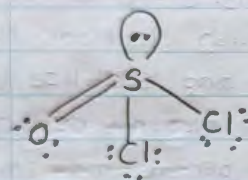
$$\frac{10}{2} = 5$$

$$\frac{8}{2} = 4$$

repulsive units

3 bond pairs

1 lone pair



Arrangement $\rightarrow$ tetrahedral.

Shape $\rightarrow$ trigonal pyramidal.

• MnO_4^-

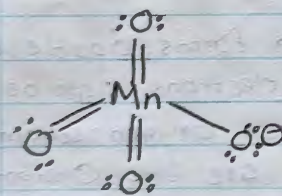
$$\text{Mn} \rightarrow 7$$

$$4\text{O} \rightarrow 8$$

$$(-) \rightarrow 1$$

$$16 - 8 = \frac{8}{2} = 4$$

removes 2e from each Oxygen atom



trigonal bipyramidal.

• SF_4

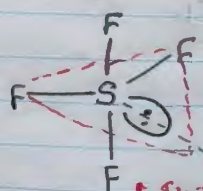
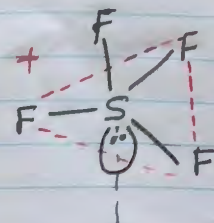
$$\text{S} \rightarrow 6$$

$$4\text{F} \rightarrow 4$$

$$\frac{10}{2} = 5$$

4 bond pairs

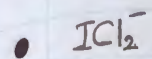
1 lone pair



In the most stable structure the lone pair is located on the plane

More stable

This is much more stable because the lone pair repulsion is minimum

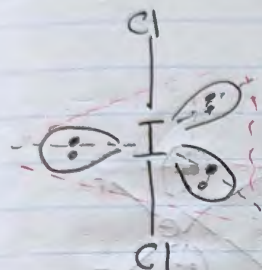


$$\begin{array}{rcl} \text{I} & \rightarrow & 7 \\ 2 \text{ Cl} & \rightarrow & 2 \\ (-) & \rightarrow & 1 \\ \hline & & 10 \end{array}$$

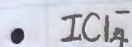
$$\frac{10}{2} \rightarrow 5 \text{ (5 repulsive units)}$$

two bonds

3 lone pairs



The most stable form



$$\begin{array}{rcl} \text{I} & \rightarrow & 7 \\ 4 \text{ Cl} & \rightarrow & 4 \\ (-) & \rightarrow & 1 \\ \hline & & 12 \end{array}$$

$$\frac{12}{2} \rightarrow 6 \text{ (6 repulsive units)}$$

In here the lone pair bond repulsion is

repulsion is

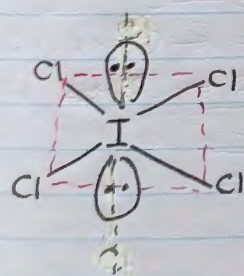
minimum when

the lone pairs

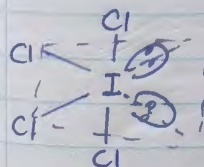
are positioned

at the opposite sides

of the octahedral plane



Shape $\rightarrow$ Square plane

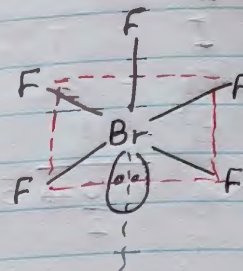


As this is highly unstable this doesn't exist.

• BrF_5

$$\begin{array}{rcl} \text{Br} & \rightarrow & 7 \\ 5\text{F} & \rightarrow & 5 \\ \hline & & 12 \\ & & \div 2 \end{array} \rightarrow 6$$

Shape
Square Pyramid



• XeF_4

$$\begin{array}{rcl} \text{Xe} & \rightarrow & 8 \\ 4\text{F} & \rightarrow & 4 \\ \hline & & 12 \\ & & \div 2 \end{array} \rightarrow 6$$

4 bond pairs
2 lone pairs

